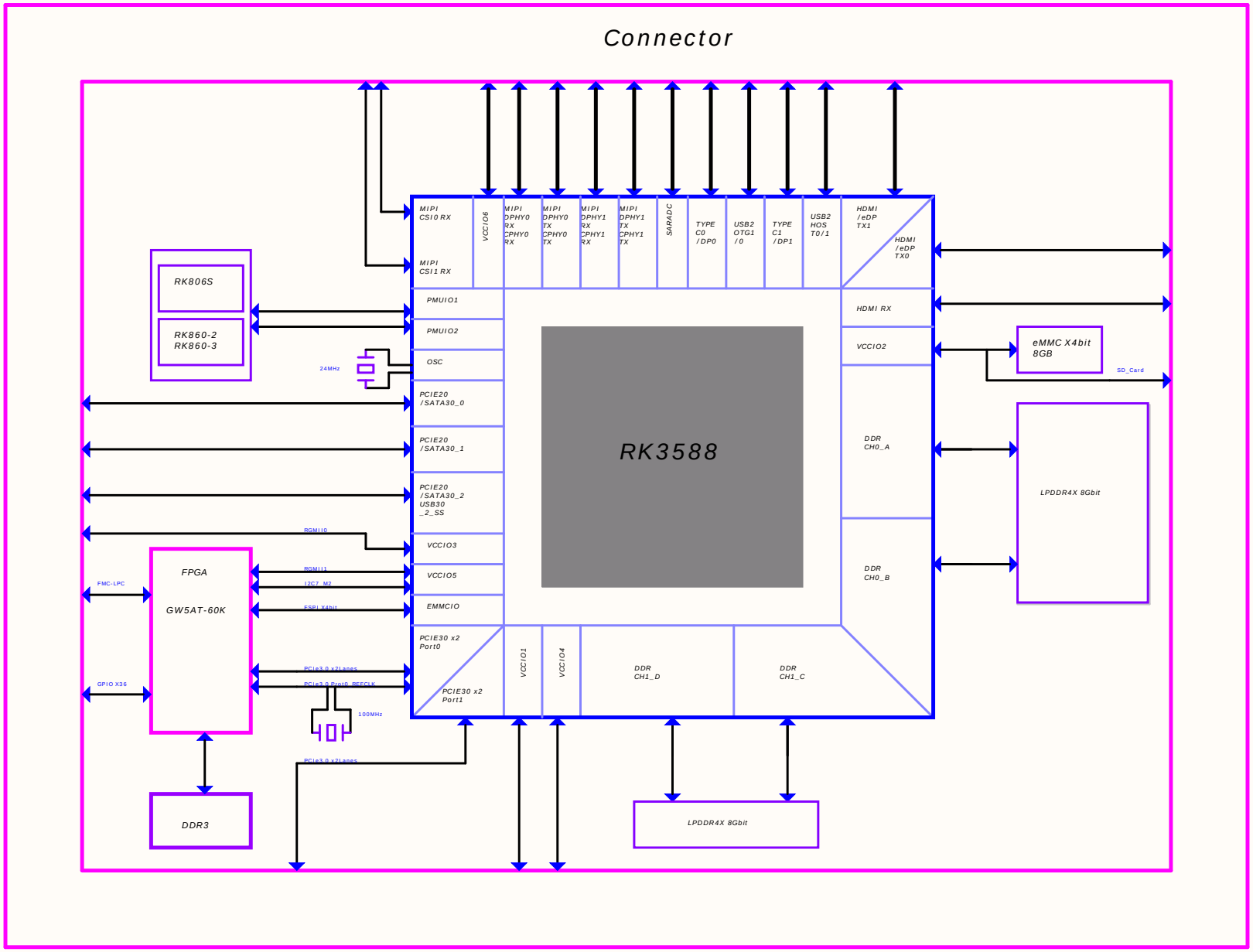
Revision History

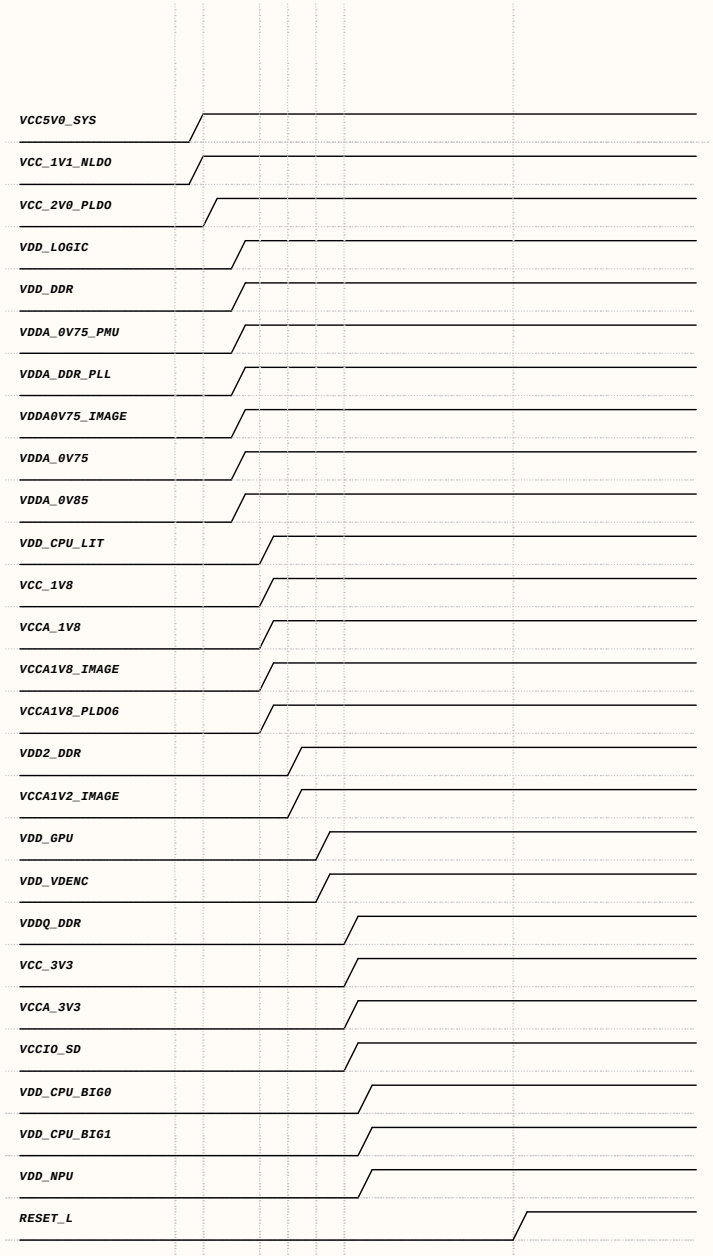
Version	Date	By	Change Dsecription	Approved
V1.0	2026-03-09 15:40	tianyu xiao		

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Size A4	Number	Revision
Date: 8-28-2026		Sheet of
File: D:\AD-PRJ\..\01_Revision History.SchDoc		Drawn By:



Title		
Size: A2	Number	Revision
Date: 8-28-2026	Sheet of	
File: D:\AD-PRJ\..\02_Block Diagram.SchDoc	Drawn By:	

Power Sequence



Power description

Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Work Voltage	Peak Current	Sleep Current
VCC5V0_SYS	RK806_BUCK1	6.5A	VDD_GPU	Slot:5	0.75V	ON	DVFS	TBD	TBD
VCC5V0_SYS	RK806_BUCK2	5A	VDD_CPU_LIT	Slot:3	0.75V	ON	DVFS	TBD	TBD
VCC5V0_SYS	RK806_BUCK3	5A	VDD_LOGIC	Slot:2	0.75V	ON	0.75V	TBD	TBD
VCC5V0_SYS	RK806_BUCK4	5A	VDD_VDENC	Slot:5	0.75V	ON	DVFS	TBD	TBD
VCC5V0_SYS	RK806_BUCK5	2.5A	VDD_DDR	Slot:2	0.85V	ON	0.85V DVFS	TBD	TBD
VCC5V0_SYS	RK806_BUCK6	2.5A	VDD2_DDR	Slot:4	ADJ FB=0.5V	ON	1.1V LPDDR4/4x	TBD	TBD
VCC5V0_SYS	RK806_BUCK7	2.5A	VCC_2V0_PLDO	Slot:1	2.0V	ON	2.0V	TBD	TBD
VCC5V0_SYS	RK806_BUCK8	2.5A	VCC_3V3	Slot:6	3.3V	ON	3.3V	TBD	TBD
VCC5V0_SYS	RK806_BUCK9	2.5A	VDDQ_DDR	Slot:6	ADJ FB=0.5V	ON	0.6V LPDDR4/4x	TBD	TBD
VCC5V0_SYS	RK806_BUCK10	2.5A	VCC_1V8	Slot:3	1.8V	ON	1.8V	TBD	TBD
VCC_2V0_PLDO	RK806_PLDO1	0.5A	VCCA_1V8	Slot:3	1.8V	ON	1.8V	TBD	TBD
	RK806_PLDO2	0.3A	VCCA1V8_IMAGE	Slot:3	1.8V	ON	1.8V	TBD	TBD
	RK806_PLDO3	0.3A	VCCA1V2_IMAGE	Slot:4	1.2V	ON	1.2V	TBD	TBD
VCC5V0_SYS	RK806_PLDO4	0.5A	VCCA_3V3	Slot:6	3.3V	ON	3.3V	TBD	TBD
	RK806_PLDO5	0.3A	VCCIO_SD	Slot:6	3.3V	ON	3.3V	TBD	TBD
VCC5V0_SYS	RK806_PLDO6	0.3A	VCCA1V8_PLDO6	Slot:3	1.8V	ON	1.8V	TBD	TBD
VCC_1V1_NLDO	RK806_NLDO1	0.3A	VDDA_0V75_PMU	Slot:2	0.75V	ON	0.75V	TBD	TBD
	RK806_NLDO2	0.3A	VDDA_DDR_PLL	Slot:2	0.85V	ON	0.85V DVFS	TBD	TBD
	RK806_NLDO3	0.5A	VDDA0V75_IMAGE	Slot:2	0.75V	ON	0.75V	**TBD**	**TBD**
VCC_1V1_NLDO	RK806_NLDO4	0.5A	VDDA_0V85	Slot:2	0.85V	ON	0.85V	TBD	TBD
	RK806_NLDO5	0.3A	VDDA_0V75	Slot:2	0.75V	ON	0.75V	TBD	TBD
VCC5V0_SYS	EXT BUCK	2A	VCC_1V1_NLDO	Slot:0A	1.1V	ON	1.1V	TBD	TBD
VCC12V_DCIN	EXT BUCK	3A	VCC5V0_SYS	Slot:0A	5.2V	ON	5.2V	TBD	TBD
VCC5V0_SYS	RK860-2	6A	VDD_CPU_BIG0	Slot:6A	0.8V	ON	DVFS	TBD	TBD
VCC5V0_SYS	RK860-3	6A	VDD_CPU_BIG1	Slot:6A	0.8V	ON	DVFS	TBD	TBD
VCC5V0_SYS	RK860-2	6A	VDD_NPU	Slot:6A	0.8V	ON	DVFS	TBD	TBD

IO Power Domain Map

IO Domain	Pin Num	Support IO Voltage	Supply Power Pin Name	Power Source	Operating Voltage
PMUIO1	Pin N28	1.8V Only	PMUIO1_1V8	VCC_1V8	1.8V
PMUIO2	Pin R27 Pin P28	1.8V or 3.3V	PMUIO2_1V8 PMUIO2	VCC_1V8 VCC_3V3	3.3V
EMMCIO	Pin V26	1.8V Only	EMMCIO_1V8	VCC_1V8	1.8V
VCCIO1	Pin G20	1.8V Only	VDDIO1_1V8	VCC_1V8	1.8V
VCCIO2	Pin AA7 Pin Y7	1.8V or 3.3V	VDDIO2_1V8 VCCIO2	VCC_1V8 VCC_3V3	3.3V
VCCIO3	Pin Y26	1.8V Only	VDDIO3_1V8	VCC_1V8	1.8V
VCCIO4	Pin H20 Pin H21	1.8V or 3.3V	VDDIO4_1V8 VCCIO4	VCC_1V8 VCC_1V8	1.8V
VCCIO5	Pin W25 Pin W26	1.8V or 3.3V	VDDIO5_1V8 VCCIO5	VCC_1V8 VCC_3V3	3.3V
VCCIO6	Pin AC25 Pin AC26	1.8V or 3.3V	VDDIO6_1V8 VCCIO6	VCC_1V8 VCC_3V3	3.3V

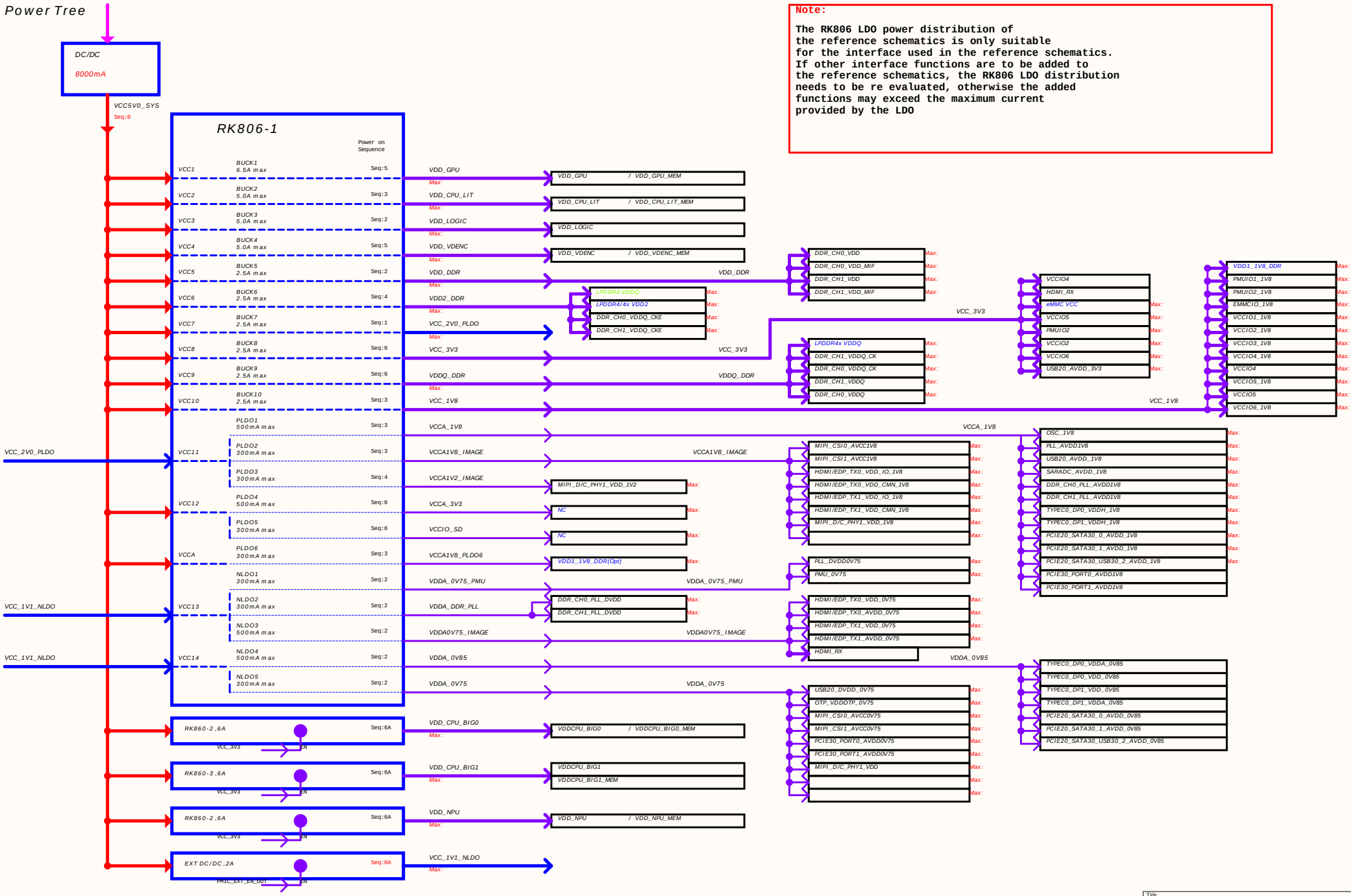
IO Type	Operating Voltage
1.8V Only	VCCIO*_1V8=1.8V
1.8V or 3.3V	VCCIO*_1V8=1.8V VCCIO*=1.8V or 3.3V

Title		
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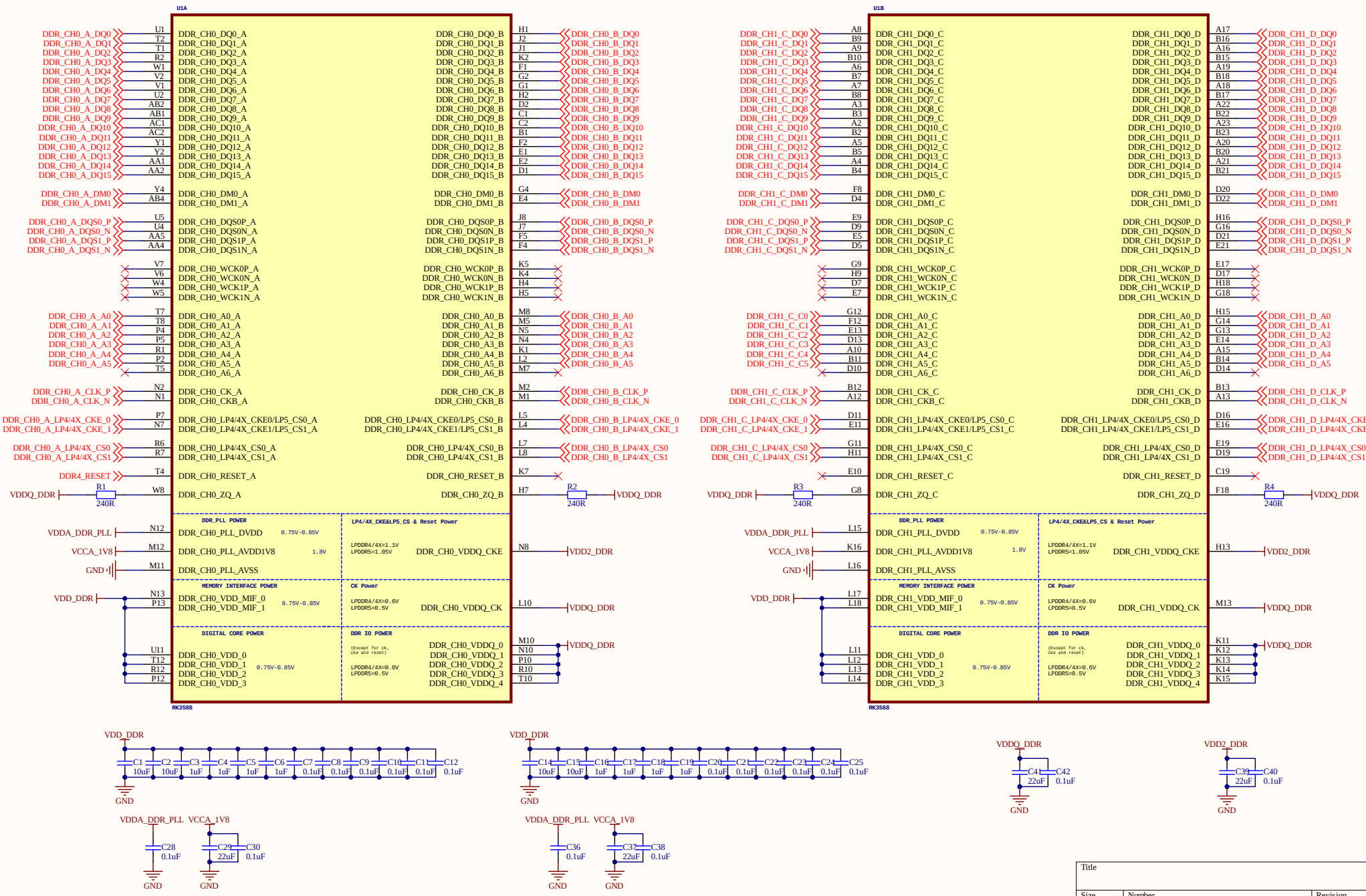
Power Tree

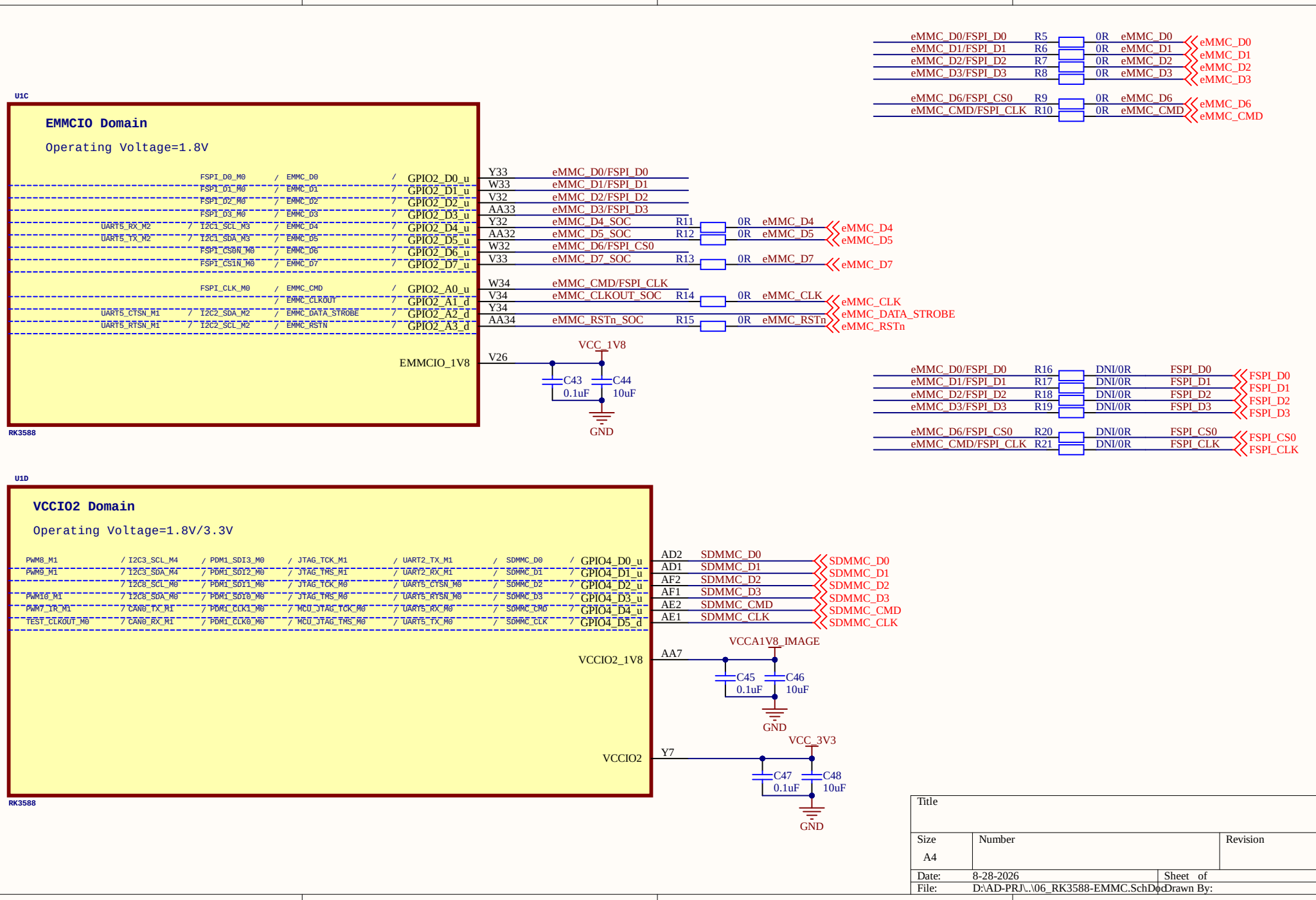
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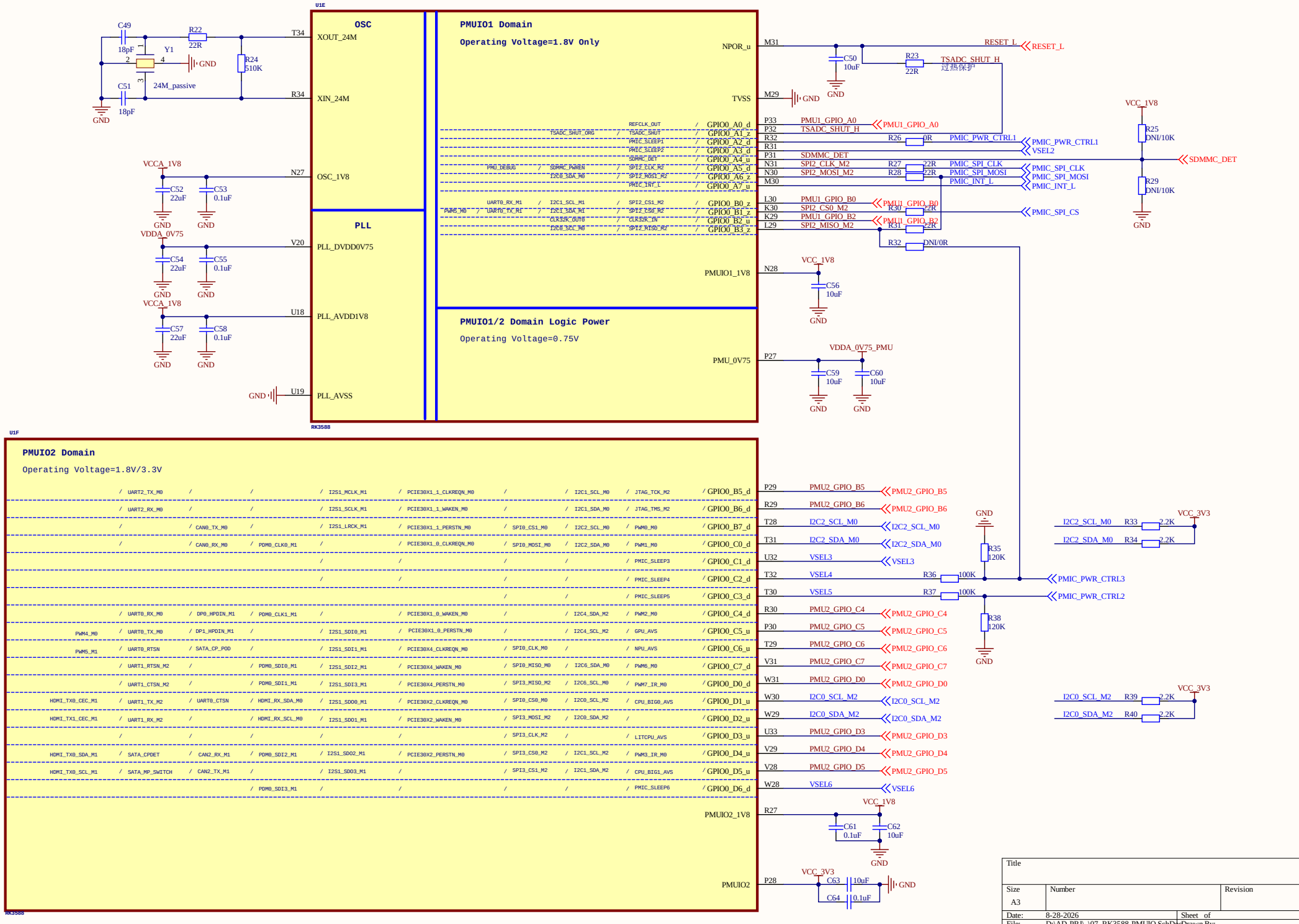
The RK806 LDO power distribution of the reference schematics is only suitable for the interface used in the reference schematics. If other interface functions are to be added to the reference schematics, the RK806 LDO distribution needs to be re evaluated, otherwise the added functions may exceed the maximum current provided by the LDO



Title		
Size: A2	Number	Revision
Date: 8-28-2026	Sheet of	
File: D:\AD-PRJ\...\04_Power Tree.SchDoc	Drawn By:	


U1A
DDR_CH0_A_DQ0
DDR_CH0_A_DQ1
DDR_CH0_A_DQ2
DDR_CH0_A_DQ3
DDR_CH0_A_DQ4
DDR_CH0_A_DQ5
DDR_CH0_A_DQ6
DDR_CH0_A_DQ7
DDR_CH0_A_DQ8
DDR_CH0_A_DQ9
DDR_CH0_A_DQ10
DDR_CH0_A_DQ11
DDR_CH0_A_DQ12
DDR_CH0_A_DQ13
DDR_CH0_A_DQ14
DDR_CH0_A_DQ15
DDR_CH0_A_DM0
DDR_CH0_A_DM1
DDR_CH0_A_DQS0_P
DDR_CH0_A_DQS0_N
DDR_CH0_A_DQS1_P
DDR_CH0_A_DQS1_N
DDR_CH0_A_A0
DDR_CH0_A_A1
DDR_CH0_A_A2
DDR_CH0_A_A3
DDR_CH0_A_A4
DDR_CH0_A_A5
DDR_CH0_A_CLK_P
DDR_CH0_A_CLK_N
DDR_CH0_A_LP4/4X_CKE_0
DDR_CH0_A_LP4/4X_CKE_1
DDR_CH0_A_LP4/4X_CS0
DDR_CH0_A_LP4/4X_CS1
DDR4_RESET
VDDQ_DDR
R1
240R
VDDA_DDR_PLL
VCCA_1V8
GND
VDD_DDR
DDR_CH0_DQ0_B
DDR_CH0_DQ1_B
DDR_CH0_DQ2_B
DDR_CH0_DQ3_B
DDR_CH0_DQ4_B
DDR_CH0_DQ5_B
DDR_CH0_DQ6_B
DDR_CH0_DQ7_B
DDR_CH0_DQ8_B
DDR_CH0_DQ9_B
DDR_CH0_DQ10_B
DDR_CH0_DQ11_B
DDR_CH0_DQ12_B
DDR_CH0_DQ13_B
DDR_CH0_DQ14_B
DDR_CH0_DQ15_B
DDR_CH0_DM0_B
DDR_CH0_DM1_B
DDR_CH0_DQS0P_B
DDR_CH0_DQS0N_B
DDR_CH0_DQS1P_B
DDR_CH0_DQS1N_B
DDR_CH0_WCK0P_B
DDR_CH0_WCK0N_B
DDR_CH0_WCK1P_B
DDR_CH0_WCK1N_B
DDR_CH0_A0_B
DDR_CH0_A1_B
DDR_CH0_A2_B
DDR_CH0_A3_B
DDR_CH0_A4_B
DDR_CH0_A5_B
DDR_CH0_A6_B
DDR_CH0_CK_B
DDR_CH0_CKB_B
DDR_CH0_LP4/4X_CKE0/LP5_CS0_B
DDR_CH0_LP4/4X_CKE1/LP5_CS1_B
DDR_CH0_LP4/4X_CS0_B
DDR_CH0_LP4/4X_CS1_B
DDR_CH0_RESET_B
DDR_CH0_ZQ_B
R2
240R
DDR_CH0_B_DQ0
DDR_CH0_B_DQ1
DDR_CH0_B_DQ2
DDR_CH0_B_DQ3
DDR_CH0_B_DQ4
DDR_CH0_B_DQ5
DDR_CH0_B_DQ6
DDR_CH0_B_DQ7
DDR_CH0_B_DQ8
DDR_CH0_B_DQ9
DDR_CH0_B_DQ10
DDR_CH0_B_DQ11
DDR_CH0_B_DQ12
DDR_CH0_B_DQ13
DDR_CH0_B_DQ14
DDR_CH0_B_DQ15
DDR_CH0_B_DM0
DDR_CH0_B_DM1
DDR_CH0_B_DQS0_P
DDR_CH0_B_DQS0_N
DDR_CH0_B_DQS1_P
DDR_CH0_B_DQS1_N
DDR_CH0_B_A0
DDR_CH0_B_A1
DDR_CH0_B_A2
DDR_CH0_B_A3
DDR_CH0_B_A4
DDR_CH0_B_A5
DDR_CH0_B_CLK_P
DDR_CH0_B_CLK_N
DDR_CH0_B_LP4/4X_CKE_0
DDR_CH0_B_LP4/4X_CKE_1
DDR_CH0_B_LP4/4X_CS0
DDR_CH0_B_LP4/4X_CS1
VDDQ_DDR
VDD2_DDR
VDDQ_DDR
VDDQ_DDR
VDDQ_DDR
DDR_PLL POWER
DDR_CH0_PLL_DVDD
0.75V-0.85V
DDR_CH0_PLL_AVDD1V8
1.8V
DDR_CH0_PLL_AVSS
LP4/4X_CKE&LP5_CS & Reset Power
LPDDR4/4X=1.1V
LPDDR5=1.05V
DDR_CH0_VDDQ_CKE
MEMORY INTERFACE POWER
DDR_CH0_VDD_MIF_0
DDR_CH0_VDD_MIF_1
0.75V-0.85V
CK Power
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH0_VDDQ_CK
DIGITAL CORE POWER
DDR_CH0_VDD_0
DDR_CH0_VDD_1
DDR_CH0_VDD_2
DDR_CH0_VDD_3
0.75V-0.85V
DDR IO POWER
(Except for ck, cke and reset)
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH0_VDDQ_0
DDR_CH0_VDDQ_1
DDR_CH0_VDDQ_2
DDR_CH0_VDDQ_3
DDR_CH0_VDDQ_4
RK3588
U1B
DDR_CH1_C_DQ0
DDR_CH1_C_DQ1
DDR_CH1_C_DQ2
DDR_CH1_C_DQ3
DDR_CH1_C_DQ4
DDR_CH1_C_DQ5
DDR_CH1_C_DQ6
DDR_CH1_C_DQ7
DDR_CH1_C_DQ8
DDR_CH1_C_DQ9
DDR_CH1_C_DQ10
DDR_CH1_C_DQ11
DDR_CH1_C_DQ12
DDR_CH1_C_DQ13
DDR_CH1_C_DQ14
DDR_CH1_C_DQ15
DDR_CH1_C_DM0
DDR_CH1_C_DM1
DDR_CH1_C_DQS0_P
DDR_CH1_C_DQS0_N
DDR_CH1_C_DQS1_P
DDR_CH1_C_DQS1_N
DDR_CH1_C_C0
DDR_CH1_C_C1
DDR_CH1_C_C2
DDR_CH1_C_C3
DDR_CH1_C_C4
DDR_CH1_C_CS
DDR_CH1_C_CLK_P
DDR_CH1_C_CLK_N
DDR_CH1_C_LP4/4X_CKE_0
DDR_CH1_C_LP4/4X_CKE_1
DDR_CH1_C_LP4/4X_CS0
DDR_CH1_C_LP4/4X_CS1
VDDQ_DDR
R3
240R
VDDA_DDR_PLL
VCCA_1V8
GND
VDD_DDR
DDR_CH1_DQ0_C
DDR_CH1_DQ1_C
DDR_CH1_DQ2_C
DDR_CH1_DQ3_C
DDR_CH1_DQ4_C
DDR_CH1_DQ5_C
DDR_CH1_DQ6_C
DDR_CH1_DQ7_C
DDR_CH1_DQ8_C
DDR_CH1_DQ9_C
DDR_CH1_DQ10_C
DDR_CH1_DQ11_C
DDR_CH1_DQ12_C
DDR_CH1_DQ13_C
DDR_CH1_DQ14_C
DDR_CH1_DQ15_C
DDR_CH1_DM0_C
DDR_CH1_DM1_C
DDR_CH1_DQS0P_C
DDR_CH1_DQS0N_C
DDR_CH1_DQS1P_C
DDR_CH1_DQS1N_C
DDR_CH1_WCK0P_C
DDR_CH1_WCK0N_C
DDR_CH1_WCK1P_C
DDR_CH1_WCK1N_C
DDR_CH1_A0_C
DDR_CH1_A1_C
DDR_CH1_A2_C
DDR_CH1_A3_C
DDR_CH1_A4_C
DDR_CH1_A5_C
DDR_CH1_A6_C
DDR_CH1_CK_C
DDR_CH1_CKB_C
DDR_CH1_LP4/4X_CKE0/LP5_CS0_C
DDR_CH1_LP4/4X_CKE1/LP5_CS1_C
DDR_CH1_LP4/4X_CS0_C
DDR_CH1_LP4/4X_CS1_C
DDR_CH1_RESET_C
DDR_CH1_ZQ_C
R4
240R
DDR_CH1_D_DQ0
DDR_CH1_D_DQ1
DDR_CH1_D_DQ2
DDR_CH1_D_DQ3
DDR_CH1_D_DQ4
DDR_CH1_D_DQ5
DDR_CH1_D_DQ6
DDR_CH1_D_DQ7
DDR_CH1_D_DQ8
DDR_CH1_D_DQ9
DDR_CH1_D_DQ10
DDR_CH1_D_DQ11
DDR_CH1_D_DQ12
DDR_CH1_D_DQ13
DDR_CH1_D_DQ14
DDR_CH1_D_DQ15
DDR_CH1_D_DM0
DDR_CH1_D_DM1
DDR_CH1_D_DQS0_P
DDR_CH1_D_DQS0_N
DDR_CH1_D_DQS1_P
DDR_CH1_D_DQS1_N
DDR_CH1_D_A0
DDR_CH1_D_A1
DDR_CH1_D_A2
DDR_CH1_D_A3
DDR_CH1_D_A4
DDR_CH1_D_A5
DDR_CH1_D_CLK_P
DDR_CH1_D_CLK_N
DDR_CH1_D_LP4/4X_CKE_0
DDR_CH1_D_LP4/4X_CKE_1
DDR_CH1_D_LP4/4X_CS0
DDR_CH1_D_LP4/4X_CS1
VDDQ_DDR
VDD2_DDR
VDDQ_DDR
VDDQ_DDR
VDDQ_DDR
DDR_PLL POWER
DDR_CH1_PLL_DVDD
0.75V-0.85V
DDR_CH1_PLL_AVDD1V8
1.8V
DDR_CH1_PLL_AVSS
LP4/4X_CKE&LP5_CS & Reset Power
LPDDR4/4X=1.1V
LPDDR5=1.05V
DDR_CH1_VDDQ_CKE
MEMORY INTERFACE POWER
DDR_CH1_VDD_MIF_0
DDR_CH1_VDD_MIF_1
0.75V-0.85V
CK Power
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH1_VDDQ_CK
DIGITAL CORE POWER
DDR_CH1_VDD_0
DDR_CH1_VDD_1
DDR_CH1_VDD_2
DDR_CH1_VDD_3
0.75V-0.85V
DDR IO POWER
(Except for ck, cke and reset)
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH1_VDDQ_0
DDR_CH1_VDDQ_1
DDR_CH1_VDDQ_2
DDR_CH1_VDDQ_3
DDR_CH1_VDDQ_4
RK3588
VDD_DDR
C1 10uF
C2 10uF
C3 1uF
C4 1uF
C5 1uF
C6 1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
C12 0.1uF
GND
VDDA_DDR_PLL
VCCA_1V8
C28 0.1uF
C29 22uF
C30 0.1uF
GND
GND
VDD_DDR
C14 10uF
C15 10uF
C16 1uF
C17 1uF
C18 1uF
C19 1uF
C20 0.1uF
C21 0.1uF
C22 0.1uF
C23 0.1uF
C24 0.1uF
C25 0.1uF
GND
VDDA_DDR_PLL
VCCA_1V8
C36 0.1uF
C37 22uF
C38 0.1uF
GND
GND
VDDQ_DDR
C41 22uF
C42 0.1uF
GND
VDD2_DDR
C39 22uF
C40 0.1uF
GND
Title
Size
A3
Number
Revision
Date:
8-28-2026
Sheet of
File:
D:\AD-PRJ\..\05_RK3588-DDRPHY.SchDoc
Drawn By:


U1C
EMMCIO Domain
Operating Voltage=1.8V
FSPI_D0_M0 / EMMC_D0 / GPIO2_D0_u
FSPI_D1_M0 / EMMC_D1 / GPIO2_D1_u
FSPI_D2_M0 / EMMC_D2 / GPIO2_D2_u
FSPI_D3_M0 / EMMC_D3 / GPIO2_D3_u
UART5_RX_M2 / I2C1_SCL_M3 / EMMC_D4 / GPIO2_D4_u
UART5_TX_M2 / I2C1_SDA_M3 / EMMC_D5 / GPIO2_D5_u
FSPI_CS0N_M0 / EMMC_D6 / GPIO2_D6_u
FSPI_CS1N_M0 / EMMC_D7 / GPIO2_D7_u
FSPI_CLK_M0 / EMMC_CMD / GPIO2_A0_u
EMMC_CLKOUT / GPIO2_A1_d
UART5_CTSN_M1 / I2C2_SDA_M2 / EMMC_DATA_STROBE / GPIO2_A2_d
UART5_RTSN_M1 / I2C2_SCL_M2 / EMMC_RSTN / GPIO2_A3_d
EMMCIO_1V8
RK3588
Y33 eMMC_D0/FSPI_D0
W33 eMMC_D1/FSPI_D1
V32 eMMC_D2/FSPI_D2
AA33 eMMC_D3/FSPI_D3
Y32 eMMC_D4_SOC R11 0R eMMC_D4 eMMC_D4
AA32 eMMC_D5_SOC R12 0R eMMC_D5 eMMC_D5
W32 eMMC_D6/FSPI_CS0
V33 eMMC_D7_SOC R13 0R eMMC_D7 eMMC_D7
W34 eMMC_CMD/FSPI_CLK
V34 eMMC_CLKOUT_SOC R14 0R eMMC_CLK eMMC_CLK
Y34
AA34 eMMC_RSTn_SOC R15 0R eMMC_RSTn eMMC_DATA_STROBE
eMMC_RSTn
V26
VCC_1V8
C43 0.1uF
C44 10uF
GND
eMMC_D0/FSPI_D0 R5 0R eMMC_D0 eMMC_D0
eMMC_D1/FSPI_D1 R6 0R eMMC_D1 eMMC_D1
eMMC_D2/FSPI_D2 R7 0R eMMC_D2 eMMC_D2
eMMC_D3/FSPI_D3 R8 0R eMMC_D3 eMMC_D3
eMMC_D6/FSPI_CS0 R9 0R eMMC_D6 eMMC_D6
eMMC_CMD/FSPI_CLK R10 0R eMMC_CMD eMMC_CMD
eMMC_D0/FSPI_D0 R16 DNI/0R FSPI_D0 FSPI_D0
eMMC_D1/FSPI_D1 R17 DNI/0R FSPI_D1 FSPI_D1
eMMC_D2/FSPI_D2 R18 DNI/0R FSPI_D2 FSPI_D2
eMMC_D3/FSPI_D3 R19 DNI/0R FSPI_D3 FSPI_D3
eMMC_D6/FSPI_CS0 R20 DNI/0R FSPI_CS0 FSPI_CS0
eMMC_CMD/FSPI_CLK R21 DNI/0R FSPI_CLK FSPI_CLK
U1D
VCCIO2 Domain
Operating Voltage=1.8V/3.3V
PWM8_M1 / I2C3_SCL_M4 / PDM1_SDI3_M0 / JTAG_TCK_M1 / UART2_TX_M1 / SDMMC_D0 / GPIO4_D0_u
PWM9_M1 / I2C3_SDA_M4 / PDM1_SDI2_M0 / JTAG_TMS_M1 / UART2_RX_M1 / SDMMC_D1 / GPIO4_D1_u
I2C8_SCL_M0 / PDM1_SDI1_M0 / JTAG_TCK_M0 / UART5_CTSN_M0 / SDMMC_D2 / GPIO4_D2_u
PWM10_M1 / I2C8_SDA_M0 / PDM1_SDI0_M0 / JTAG_TMS_M0 / UART5_RTSN_M0 / SDMMC_D3 / GPIO4_D3_u
PWM7_IR_M1 / CAN0_TX_M1 / PDM1_CLK1_M0 / MCU_JTAG_TCK_M0 / UART5_RX_M0 / SDMMC_CMD / GPIO4_D4_u
TEST_CLKOUT_M0 / CAN0_RX_M1 / PDM1_CLK0_M0 / MCU_JTAG_TMS_M0 / UART5_TX_M0 / SDMMC_CLK / GPIO4_D5_d
VCCIO2_1V8
VCCIO2
RK3588
AD2 SDMMC_D0 SDMMC_D0
AD1 SDMMC_D1 SDMMC_D1
AF2 SDMMC_D2 SDMMC_D2
AF1 SDMMC_D3 SDMMC_D3
AE2 SDMMC_CMD SDMMC_CMD
AE1 SDMMC_CLK SDMMC_CLK
VCCA1V8_IMAGE
AA7
C45 0.1uF
C46 10uF
GND
VCC_3V3
Y7
C47 0.1uF
C48 10uF
GND
Title
Size A4
Number
Revision
Date: 8-28-2026
Sheet of
File: D:\AD-PRJ\..\06_RK3588-EMMC.SchDoc
Drawn By:


U1E
OSC
XOUT_24M
XIN_24M
PLL
PLL_DVDD0V75
PLL_AVDD1V8
PLL_AVSS
OSC_1V8
RK3588
PMUIO1 Domain
Operating Voltage=1.8V Only
NPOR_u
TVSS
PMUIO1_1V8
PMUIO1/2 Domain Logic Power
Operating Voltage=0.75V
PMU_0V75
U1F
PMUIO2 Domain
Operating Voltage=1.8V/3.3V
PMUIO2_1V8
PMUIO2
C49
18pF
R22
22R
Y1
24M_passive
R24
510K
C51
18pF
C52
22uF
C53
0.1uF
C54
22uF
C55
0.1uF
C57
22uF
C58
0.1uF
C50
10uF
R23
22R
TSADC_SHUT_H
过热保护
C56
10uF
C59
10uF
C60
10uF
C61
0.1uF
C62
10uF
C63
10uF
C64
0.1uF
RESET_L
PMIC_PWR_CTRL1
VSEL2
SDMMC_DET
PMIC_SPI_CLK
PMIC_SPI_MOSI
PMIC_INT_L
PMIC_SPI_CS
PMIC_PWR_CTRL3
PMIC_PWR_CTRL2
R25
DNI/10K
R29
DNI/10K
R32
DNI/0R
R35
120K
R36
100K
R37
100K
R38
120K
R33
2.2K
R34
2.2K
R39
2.2K
R40
2.2K
VCC_1V8
VCC_3V3
VCCA_1V8
VDDA_0V75
VDDA_0V75_PMU
GND
PMU2_GPIO_B5
PMU2_GPIO_B6
I2C2_SCL_M0
I2C2_SDA_M0
VSEL3
VSEL4
VSEL5
PMU2_GPIO_C4
PMU2_GPIO_C5
PMU2_GPIO_C6
PMU2_GPIO_C7
PMU2_GPIO_D0
I2C0_SCL_M2
I2C0_SDA_M2
PMU2_GPIO_D3
PMU2_GPIO_D4
PMU2_GPIO_D5
VSEL6
PMU1_GPIO_A0
TSADC_SHUT_H
PMU1_GPIO_B0
PMU1_GPIO_B2
Title
Size
A3
Number
Revision
Date: 8-28-2026
Sheet of
File: D:\AD-PRJ\..\07_RK3588-PMUIO.SchDoc
Drawn By:

U1G

VCCIO1 Domain

Operating Voltage=1.8V

SPI4_MISO_M0	UART3_RX_M0		I2C3_SDA_M0			GPIO1_C0_z
SPI4_MOSI_M0	UART3_TX_M0		I2C3_SCL_M0			GPIO1_C1_z
SPI4_CLK_M0	UART3_RTSN	PWM3_IR_M2	I2C6_SDA_M1		I2S0_MCLK	GPIO1_C2_d
SPI4_CS0_M0	UART3_CTSN	PWM7_IR_M2	I2C6_SCL_M1		I2S0_SCLK	GPIO1_C3_d
SPI4_CS1_M0		PWM11_IR_M2	I2C2_SDA_M3	PDM0_CLK1_M0		GPIO1_C4_d
	UART4_RTSN		I2C2_SCL_M3		I2S0_LRCK	GPIO1_C5_d
		PWM15_IR_M2	I2C4_SDA_M4	PDM0_CLK0_M0		GPIO1_C6_d
	UART4_CTSN		I2C4_SCL_M4		I2S0_SDO0	GPIO1_C7_d
SPI1_MISO_M2	UART6_TX_M2		I2C7_SCL_M0		I2S0_SDO1	GPIO1_D0_d
SPI1_MOSI_M2	UART6_RX_M2		I2C7_SDA_M0	PDM0_SDI1_M0	I2S0_SDO2/I2S0_SDI3	GPIO1_D1_d
SPI1_CLK_M2	UART4_TX_M0	PWM0_M1	I2C1_SCL_M4	PDM0_SDI2_M0	I2S0_SDO3/I2S0_SDI2	GPIO1_D2_d
SPI1_CS0_M2	UART4_RX_M0	PWM1_M1	I2C1_SDA_M4	PDM0_SDI3_M0	I2S0_SDI1	GPIO1_D3_d
					I2S0_SDI0	GPIO1_D4_d
SPI1_CS1_M2				PDM0_SDI0_M0		GPIO1_D5_d
						VCCIO1_1V8

RK3588

G29 B1_GPIO1_C0 → B1_GPIO1_C0
G27 B1_GPIO1_C1 → B1_GPIO1_C1
F30 B1_GPIO1_C2 → B1_GPIO1_C2
E31 B1_GPIO1_C3 → B1_GPIO1_C3
E30 B1_GPIO1_C4 → B1_GPIO1_C4
D30 B1_GPIO1_C5 → B1_GPIO1_C5
D29 B1_GPIO1_C6 → B1_GPIO1_C6
E29 B1_GPIO1_C7 → B1_GPIO1_C7

F26 B1_GPIO1_D0 → B1_GPIO1_D0
F27 B1_GPIO1_D1 → B1_GPIO1_D1
F28 B1_GPIO1_D2 → B1_GPIO1_D2
E28 B1_GPIO1_D3 → B1_GPIO1_D3
D28 B1_GPIO1_D4 → B1_GPIO1_D4
G26 B1_GPIO1_D5 → B1_GPIO1_D5

G20 → VCC_1V8, C65 0.1uF, C66 10uF, GND

U1H

VCCIO3 Domain

Operating Voltage=1.8V

	UART6_RX_M0	FSPI_D0_M1	SDIO_D0_M0	GMAC0_RXD2		GPIO2_A6_u
	UART6_TX_M0	FSPI_D1_M1	SDIO_D1_M0	GMAC0_RXD3		GPIO2_A7_u
I2C8_SCL_M1	UART6_RTSN_M0	FSPI_D2_M1	SDIO_D2_M0	GMAC0_RXCLK		GPIO2_B0_u
I2C8_SDA_M1	UART6_CTSN_M0	FSPI_D3_M1	SDIO_D3_M0	GMAC0_TXD2		GPIO2_B1_u
I2C3_SCL_M3			SDIO_CMD_M0	GMAC0_TXD3		GPIO2_B2_u
I2C3_SDA_M3		FSPI_CLK_M1	SDIO_CLK_M0	GMAC0_TXCLK		GPIO2_B3_d
I2C4_SDA_M1	UART7_RX_M0	FSPI_CS0N_M1	HDMI_TX1_SDA_M0	GMAC0_PTP_REFCLK		GPIO2_B4_u
I2C4_SCL_M1	UART7_TX_M0	FSPI_CS1N_M1	HDMI_TX1_SCL_M0	GMAC0_PPSTRIG		GPIO2_B5_u
I2C5_SCL_M4	UART1_RX_M0		I2S2_MCLK_M0	GMAC0_TXD0		GPIO2_B6_d
I2C5_SDA_M4	UART1_TX_M0		I2S2_SCLK_M0	GMAC0_TXD1		GPIO2_B7_d
I2C2_SDA_M1	UART1_RTSN_M0	SPI1_CLK_M0	I2S2_LRCK_M0	GMAC0_TXEN		GPIO2_C0_d
I2C2_SCL_M1	UART1_CTSN_M0	SPI1_MISO_M0		GMAC0_RXD0		GPIO2_C1_d
I2C6_SDA_M2	UART9_TX_M0	SPI1_MOSI_M0		GMAC0_RXD1		GPIO2_C2_d
I2C6_SCL_M2		SPI1_CS0_M0	I2S2_SDI_M0	ETH0_REFCLKO_25M		GPIO2_C3_d
TEST_CLKOUT_M1	UART9_RX_M0	SPI1_CS1_M0	HDMI_TX1_CEC_M0	GMAC0_PPSCLK		GPIO2_C4_d
				CLK32K_OUT1		GPIO2_C5_d
	UART7_RTSN_M0	SPI3_CS0_M0	PWM2_M2	GMAC0_RXDV_CRS		GPIO4_C2_d
I2C7_SCL_M1	PWM4_M1	SPI3_CS1_M0	I2S2_SDO_M0	GMAC0_MCLKINOUT		GPIO4_C3_d
I2C7_SDA_M1	UART9_RTSN_M0	SPI3_MISO_M0	PWM5_M2	GMAC0_MDC		GPIO4_C4_d
I2C0_SCL_M1	UART9_CTSN_M0	SPI3_MOSI_M0	PWM6_M2	GMAC0_MDIO		GPIO4_C5_d
I2C0_SDA_M1	UART7_CTSN_M0	SPI3_CLK_M0	PWM7_IR_M3	GMAC0_TXER		GPIO4_C6_d
						VCCIO3_1V8

RK3588

AC32 ENET0_RX_DATA2 → ENET0_RX_DATA2
AC31 ENET0_RX_DATA3 → ENET0_RX_DATA3

AE32 ENET0_RX_CLK → ENET0_RX_CLK
AC33 ENET0_TX_DATA2 → ENET0_TX_DATA2
AC34 ENET0_TX_DATA3 → ENET0_TX_DATA3
AE33 ENET0_GTX_CLK → ENET0_GTX_CLK
AB31 B3_GPIO2_B4 → B3_GPIO2_B4
AB30 B3_GPIO2_B5 → B3_GPIO2_B5
AD33 ENET0_TX_DATA0 → ENET0_TX_DATA0
AD34 ENET0_TX_DATA1 → ENET0_TX_DATA1

AE34 ENET0_TX_EN → ENET0_TX_EN
AD32 ENET0_RX_DATA0 → ENET0_RX_DATA0
AD31 ENET0_RX_DATA1 → ENET0_RX_DATA1
AD30 ETH0_REFCLKO25M → ETH0_REFCLKO25M
AC30 B3_GPIO2_C4 → B3_GPIO2_C4
AE30 B3_GPIO2_C5 → B3_GPIO2_C5

AE31 ENET0_RX_DV → ENET0_RX_DV
AF34 B3_GPIO4_C3 → B3_GPIO4_C3
AB34 ENET0_MDC → ENET0_MDC
AB33 ENET0_MDIO → ENET0_MDIO
AF33 B3_GPIO4_C6 → B3_GPIO4_C6

Y26 → VCC_1V8, C67 0.1uF, C68 10uF, GND

Title		
Size A4	Number	Revision
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File: D:\AD-PRJ\..\08_RK3588-I2S&ETH0.SchDoc	Drawn By:	

U1J

VCCIO5 Domain

Operating Voltage=1.8V/3.3V

PWM10_M0	SPI4_MISO_M1		I2C6_SDA_M4	FSPI_D0_M2	I2S3_MCLK	SDIO_D0_M1	GMAC1_TXD2	GPIO3_A0_u
AUDDSM_LN	SPI4_MOSI_M1	PWM11_IR_M0	I2C6_SCL_M4	FSPI_D1_M2	I2S3_SCLK	SDIO_D1_M1	GMAC1_TXD3	GPIO3_A1_u
AUDDSM_LP	SPI4_CLK_M1	UART8_TX_M1		FSPI_D2_M2	I2S3_LRCK	SDIO_D2_M1	GMAC1_RXD2	GPIO3_A2_u
AUDDSM_RN	SPI4_CS0_M1	UART8_RX_M1		FSPI_D3_M2	I2S3_SDO	SDIO_D3_M1	GMAC1_RXD3	GPIO3_A3_u
AUDDSM_RP	SPI4_CS1_M1	UART8_RTSN_M1			I2S3_SDI	SDIO_CMD_M1	GMAC1_TXCLK	GPIO3_A4_d
	MIPI_CAMERA0_CLK_M1	UART8_CTSN_M1	I2C4_SDA_M0	FSPI_CLK_M2		SDIO_CLK_M1	GMAC1_RXCLK	GPIO3_A5_d
	MIPI_CAMERA1_CLK_M1		I2C4_SCL_M0				ETH1_REFCLKO_25M	GPIO3_A6_d
PWM8_M0	MIPI_CAMERA2_CLK_M1						GMAC1_RXD0	GPIO3_A7_u
PWM9_M0	MIPI_CAMERA3_CLK_M1						GMAC1_RXD1	GPIO3_B0_u
PWM2_M1	MIPI_CAMERA4_CLK_M1	UART2_TX_M2					GMAC1_RXDV_CRS	GPIO3_B1_d
PWM3_IR_M1		UART2_RX_M2			I2S2_SDI_M1		GMAC1_TXER	GPIO3_B2_d
		UART2_RTSN			I2S2_SDO_M1		GMAC1_TXD0	GPIO3_B3_u
		UART2_CTSN			I2S2_MCLK_M1		GMAC1_TXD1	GPIO3_B4_u
PWM12_M0	CAN1_RX_M0	UART3_TX_M1			I2S2_SCLK_M1		GMAC1_TXEN	GPIO3_B5_u
PWM13_M0	CAN1_TX_M0	UART3_RX_M1			I2S2_LRCK_M1		GMAC1_MCLKINOUT	GPIO3_B6_d
			I2C3_SCL_M1	SPI1_MOSI_M1	HDMI_TX1_HPD_M1		GMAC1_PTP_REF_CLK	GPIO3_B7_d
		UART7_TX_M1	I2C3_SDA_M1	SPI1_MISO_M1			GMAC1_PPSTRIG	GPIO3_C0_d
	PCIE30X2_BUTTON_RSTN	UART7_RX_M1		SPI1_CLK_M1			GMAC1_PPSCLK	GPIO3_C1_d
PWM14_M0		UART7_RTSN_M1	I2C8_SCL_M4	SPI1_CS0_M1		MIPI_TE0	GMAC1_MDC	GPIO3_C2_d
PWM15_IR_M0		UART7_CTSN_M1	I2C8_SDA_M4	SPI1_CS1_M1		MIPI_TE1	GMAC1_MDIO	GPIO3_C3_d
CAN2_RX_M0	PCIE30X4_CLKREQN_M2	UART5_TX_M1	FSPI_CS0N_M2	SPI3_CS0_M3	HDMI_TX1_CEC_M2		CIF_D8	GPIO3_C4_u
CAN2_TX_M0	PCIE30X4_WAKEN_M2	UART5_RX_M1	FSPI_CS1N_M2	SPI3_CS1_M3	HDMI_TX1_SDA_M1		CIF_D9	GPIO3_C5_u
	PCIE30X4_PERSTN_M2			SPI3_MISO_M3	HDMI_TX1_SCL_M1		CIF_D10	GPIO3_C6_u
	PCIE20X1_2_CLKREQN_M0		I2C5_SCL_M0	SPI3_MOSI_M3	HDMI_TX0_SCL_M2		CIF_D11	GPIO3_C7_u
PWM8_M2	PCIE20X1_2_WAKEN_M0	UART4_RX_M1	I2C5_SDA_M0	SPI3_CLK_M3	HDMI_TX0_SDA_M2		CIF_D12	GPIO3_D0_u
PWM9_M2	PCIE20X1_2_PERSTN_M0	UART4_TX_M1		SPI0_MISO_M3	HDMI_RX_CEC_M1		CIF_D13	GPIO3_D1_d
	PCIE30X2_CLKREQN_M2	UART9_RTSN_M2	I2C7_SCL_M2	SPI0_MOSI_M3	HDMI_RX_SCL_M1		CIF_D14	GPIO3_D2_d
PWM10_M2	PCIE30X2_WAKEN_M2	UART9_CTSN_M2	I2C7_SDA_M2	SPI0_CLK_M3	HDMI_RX_SDA_M1		CIF_D15	GPIO3_D3_d
	PCIE30X2_PERSTN_M2	UART9_RX_M2		SPI0_CS0_M3	HDMI_RX_HPDOUT_M1	HDMI_TX0_HPD_M1	MCU_JTAG_TCK_M1	GPIO3_D4_d
PWM11_IR_M3	PCIE30X4_BUTTON_RSTN	UART9_TX_M2		SPI0_CS1_M3	DP1_HPDIN_M0		MCU_JTAG_TMS_M1	GPIO3_D5_d

VCCIO5_1V8 W25

VCCIO5 W26

RK3588

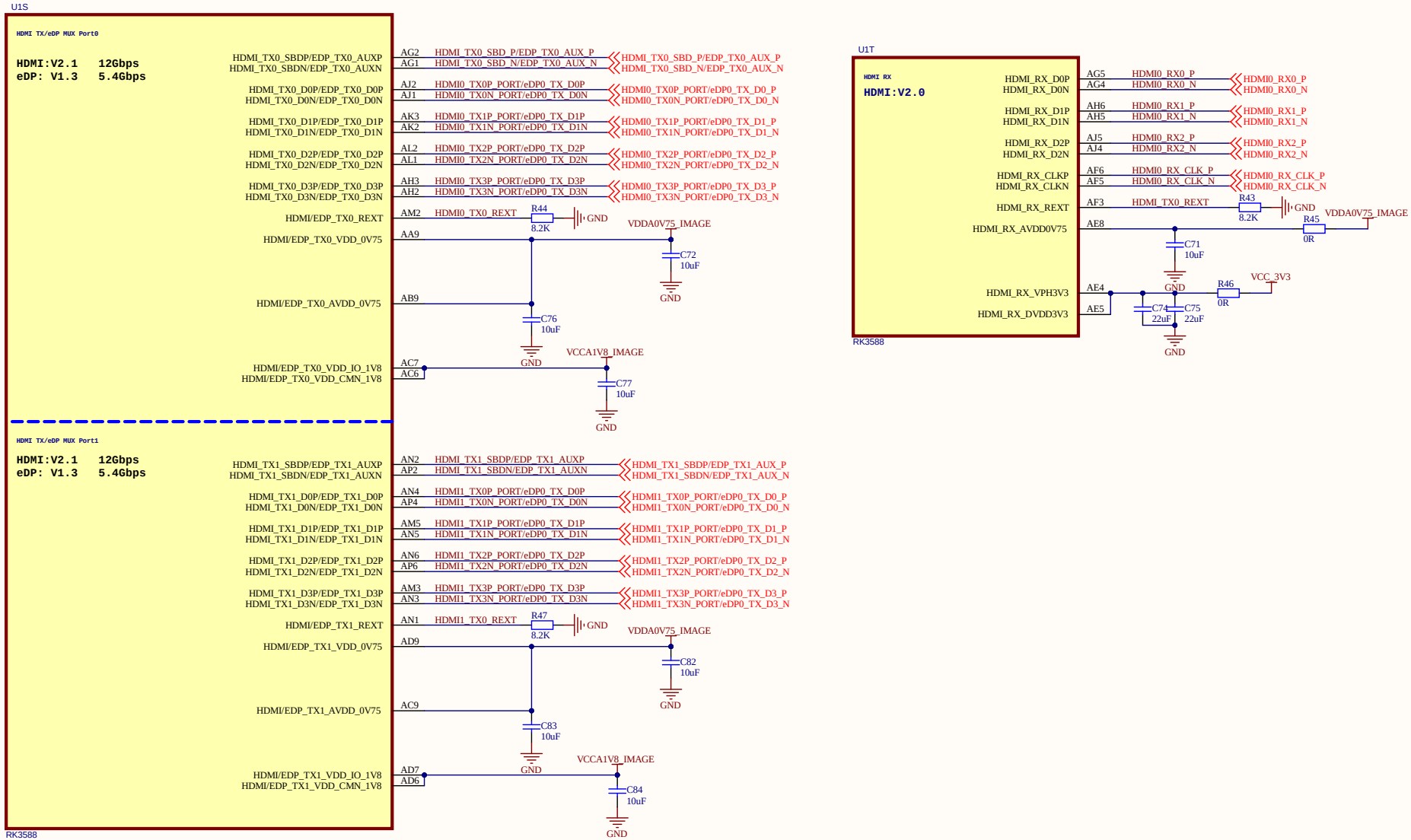
AA29 ENETI_TX_DATA2 — ENETI_TX_DATA2
AA30 ENETI_TX_DATA3 — ENETI_TX_DATA3
AD27 ENETI_RX_DATA2 — ENETI_RX_DATA2
AE27 ENETI_RX_DATA3 — ENETI_RX_DATA3
AD28 ENETI_GTX_CLK — ENETI_GTX_CLK
AH30 ENETI_RX_CLK — ENETI_RX_CLK
AH27 ETH1_REFCLKO25M — ETH1_REFCLKO25M
AG29 ENETI_RX_DATA0 — ENETI_RX_DATA0
AG28 ENETI_RX_DATA1 — ENETI_RX_DATA1
AH29 ENETI_RX_DV — ENETI_RX_DV
AE28 B5_GPIO3_B2 — B5_GPIO3_B2
AC28 ENETI_TX_DATA0 — ENETI_TX_DATA0
AC29 ENETI_TX_DATA1 — ENETI_TX_DATA1
AD29 ENETI_TX_EN — ENETI_TX_EN
AE29 B5_GPIO3_B6 — B5_GPIO3_B6
AA28 B5_GPIO3_B7 — B5_GPIO3_B7
Y29 B5_GPIO3_C0 — B5_GPIO3_C0
Y27 B5_GPIO3_C1 — B5_GPIO3_C1
Y31 ENETI_MDC — ENETI_MDC
Y30 ENETI_MDIO — ENETI_MDIO
AH26 B5_GPIO3_C4 — B5_GPIO3_C4
AH25 B5_GPIO3_C5 — B5_GPIO3_C5
AG26 B5_GPIO3_C6 — B5_GPIO3_C6
AJ24 B5_GPIO3_C7 — B5_GPIO3_C7
AH24 B5_GPIO3_D0 — B5_GPIO3_D0
AG23 B5_GPIO3_D1 — B5_GPIO3_D1
AG25 I2C7_SCL_M2 — I2C7_SCL_M2
AG24 I2C7_SDA_M2 — I2C7_SDA_M2
AA27 B5_GPIO3_D4 — B5_GPIO3_D4
AB28 B5_GPIO3_D5 — B5_GPIO3_D5

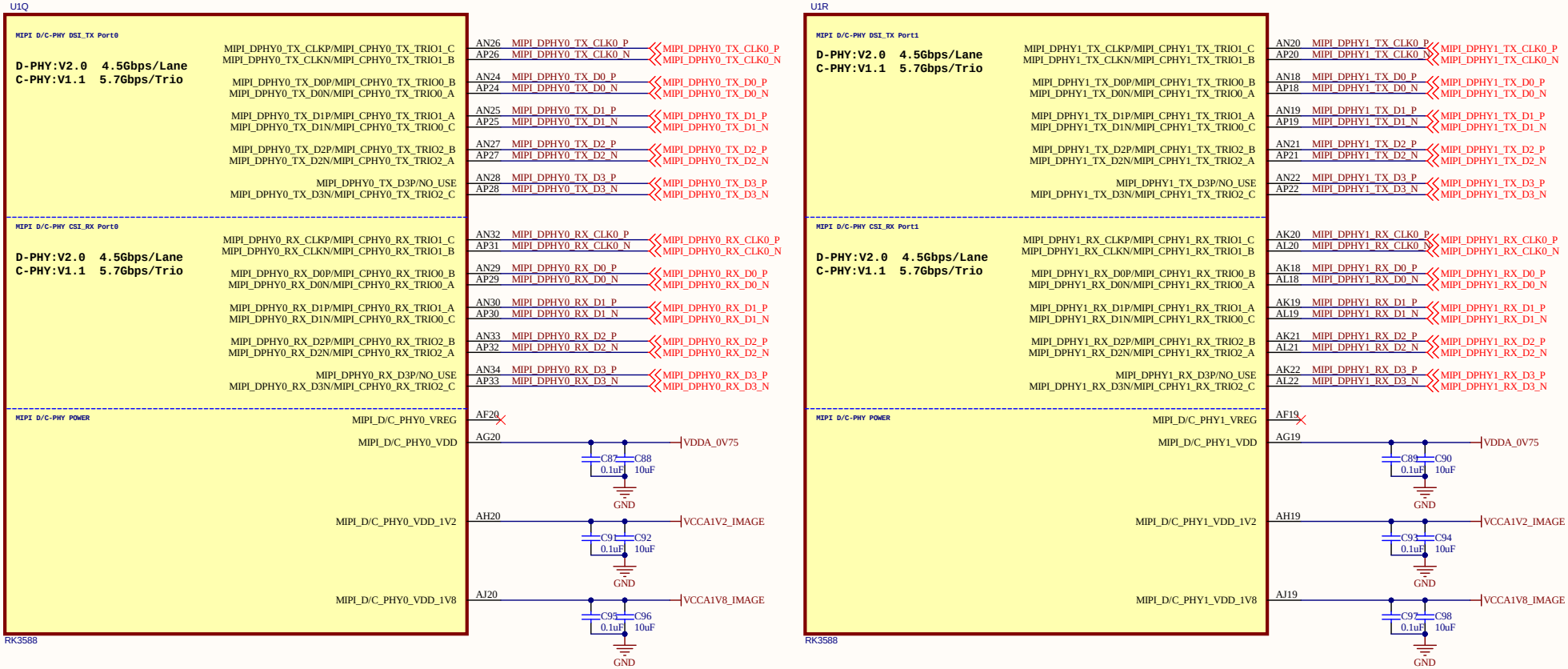
VCC_1V8
C69 10uF
GND

RK_VCCIO5
C70 10uF
GND
FB13 VCC_1V8 BLM18KG121TN1D
FB29 VCC_3V3 DNI/BLM18KG121TN1D

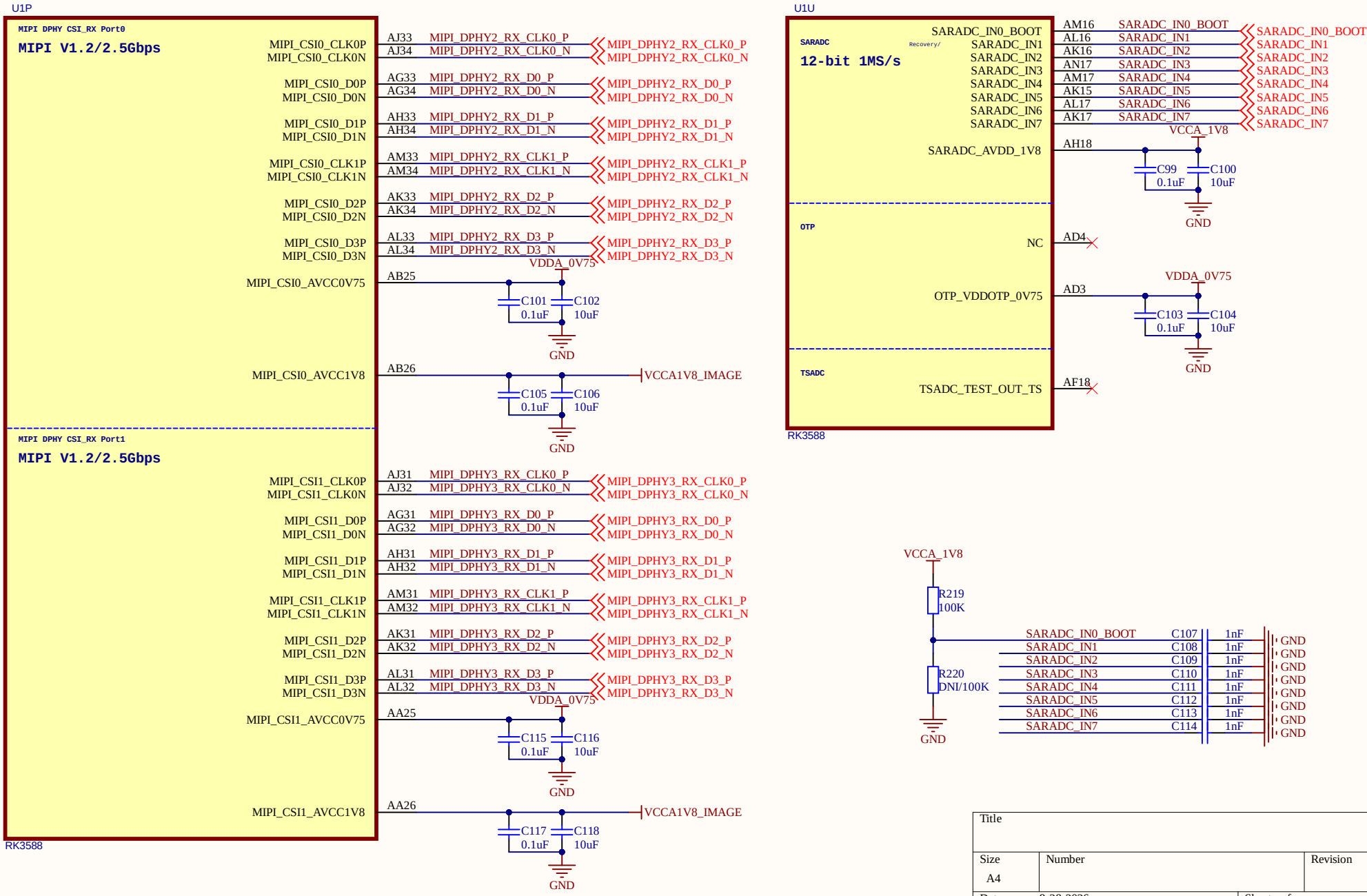
I2C7_SCL_M2 R41 2.2K RK_VCCIO5
I2C7_SDA_M2 R42 2.2K

Title		
Size A3	Number	Revision
Date: 8-28-2026	Sheet of	
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U1S
HDMI TX/eDP MUX Port0
HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps
HDMI_TX0_SBDP/EDP_TX0_AUXP
HDMI_TX0_SBDN/EDP_TX0_AUXN
HDMI_TX0_D0P/EDP_TX0_D0P
HDMI_TX0_D0N/EDP_TX0_D0N
HDMI_TX0_D1P/EDP_TX0_D1P
HDMI_TX0_D1N/EDP_TX0_D1N
HDMI_TX0_D2P/EDP_TX0_D2P
HDMI_TX0_D2N/EDP_TX0_D2N
HDMI_TX0_D3P/EDP_TX0_D3P
HDMI_TX0_D3N/EDP_TX0_D3N
HDMI/EDP_TX0_REXT
HDMI/EDP_TX0_VDD_0V75
HDMI/EDP_TX0_AVDD_0V75
HDMI/EDP_TX0_VDD_IO_1V8
HDMI/EDP_TX0_VDD_CMN_1V8
AG2 HDMI_TX0_SBD_P/EDP_TX0_AUX_P
AG1 HDMI_TX0_SBD_N/EDP_TX0_AUX_N
HDMI_TX0_SBD_P/EDP_TX0_AUX_P
HDMI_TX0_SBD_N/EDP_TX0_AUX_N
AJ2 HDMI0_TX0P_PORT/eDP0_TX_D0P
AJ1 HDMI0_TX0N_PORT/eDP0_TX_D0N
HDMI0_TX0P_PORT/eDP0_TX_D0_P
HDMI0_TX0N_PORT/eDP0_TX_D0_N
AK3 HDMI0_TX1P_PORT/eDP0_TX_D1P
AK2 HDMI0_TX1N_PORT/eDP0_TX_D1N
HDMI0_TX1P_PORT/eDP0_TX_D1_P
HDMI0_TX1N_PORT/eDP0_TX_D1_N
AL2 HDMI0_TX2P_PORT/eDP0_TX_D2P
AL1 HDMI0_TX2N_PORT/eDP0_TX_D2N
HDMI0_TX2P_PORT/eDP0_TX_D2_P
HDMI0_TX2N_PORT/eDP0_TX_D2_N
AH3 HDMI0_TX3P_PORT/eDP0_TX_D3P
AH2 HDMI0_TX3N_PORT/eDP0_TX_D3N
HDMI0_TX3P_PORT/eDP0_TX_D3_P
HDMI0_TX3N_PORT/eDP0_TX_D3_N
AM2 HDMI0_TX0_REXT
R44
8.2K
GND
VDDA0V75_IMAGE
AA9
C72
10uF
GND
AB9
C76
10uF
GND
VCCA1V8_IMAGE
AC7
AC6
C77
10uF
GND
HDMI TX/eDP MUX Port1
HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps
HDMI_TX1_SBDP/EDP_TX1_AUXP
HDMI_TX1_SBDN/EDP_TX1_AUXN
HDMI_TX1_D0P/EDP_TX1_D0P
HDMI_TX1_D0N/EDP_TX1_D0N
HDMI_TX1_D1P/EDP_TX1_D1P
HDMI_TX1_D1N/EDP_TX1_D1N
HDMI_TX1_D2P/EDP_TX1_D2P
HDMI_TX1_D2N/EDP_TX1_D2N
HDMI_TX1_D3P/EDP_TX1_D3P
HDMI_TX1_D3N/EDP_TX1_D3N
HDMI/EDP_TX1_REXT
HDMI/EDP_TX1_VDD_0V75
HDMI/EDP_TX1_AVDD_0V75
HDMI/EDP_TX1_VDD_IO_1V8
HDMI/EDP_TX1_VDD_CMN_1V8
AN2 HDMI_TX1_SBDP/EDP_TX1_AUXP
AP2 HDMI_TX1_SBDN/EDP_TX1_AUXN
HDMI_TX1_SBDP/EDP_TX1_AUX_P
HDMI_TX1_SBDN/EDP_TX1_AUX_N
AN4 HDMI1_TX0P_PORT/eDP0_TX_D0P
AP4 HDMI1_TX0N_PORT/eDP0_TX_D0N
HDMI1_TX0P_PORT/eDP0_TX_D0_P
HDMI1_TX0N_PORT/eDP0_TX_D0_N
AM5 HDMI1_TX1P_PORT/eDP0_TX_D1P
AN5 HDMI1_TX1N_PORT/eDP0_TX_D1N
HDMI1_TX1P_PORT/eDP0_TX_D1_P
HDMI1_TX1N_PORT/eDP0_TX_D1_N
AN6 HDMI1_TX2P_PORT/eDP0_TX_D2P
AP6 HDMI1_TX2N_PORT/eDP0_TX_D2N
HDMI1_TX2P_PORT/eDP0_TX_D2_P
HDMI1_TX2N_PORT/eDP0_TX_D2_N
AM3 HDMI1_TX3P_PORT/eDP0_TX_D3P
AN3 HDMI1_TX3N_PORT/eDP0_TX_D3N
HDMI1_TX3P_PORT/eDP0_TX_D3_P
HDMI1_TX3N_PORT/eDP0_TX_D3_N
AN1 HDMI1_TX0_REXT
R47
8.2K
GND
VDDA0V75_IMAGE
AD9
C82
10uF
GND
AC9
C83
10uF
GND
VCCA1V8_IMAGE
AD7
AD6
C84
10uF
GND
RK3588
U1T
HDMI RX
HDMI:V2.0
HDMI_RX_D0P
HDMI_RX_D0N
HDMI_RX_D1P
HDMI_RX_D1N
HDMI_RX_D2P
HDMI_RX_D2N
HDMI_RX_CLKP
HDMI_RX_CLKN
HDMI_RX_REXT
HDMI_RX_AVDD0V75
HDMI_RX_VPH3V3
HDMI_RX_DVDD3V3
AG5 HDMI0_RX0_P
AG4 HDMI0_RX0_N
HDMI0_RX0_P
HDMI0_RX0_N
AH6 HDMI0_RX1_P
AH5 HDMI0_RX1_N
HDMI0_RX1_P
HDMI0_RX1_N
AJ5 HDMI0_RX2_P
AJ4 HDMI0_RX2_N
HDMI0_RX2_P
HDMI0_RX2_N
AF6 HDMI0_RX_CLK_P
AF5 HDMI0_RX_CLK_N
HDMI0_RX_CLK_P
HDMI0_RX_CLK_N
AF3 HDMI_TX0_REXT
R43
8.2K
GND
R45
0R
VDDA0V75_IMAGE
AE8
C71
10uF
GND
VCC_3V3
AE4
AE5
R46
0R
C74
22uF
C75
22uF
GND
RK3588
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Number
Revision
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U1P
MIPI DPHY CSI_RX Port0
MIPI V1.2/2.5Gbps
MIPI_CSI0_CLK0P
MIPI_CSI0_CLK0N
MIPI_CSI0_D0P
MIPI_CSI0_D0N
MIPI_CSI0_D1P
MIPI_CSI0_D1N
MIPI_CSI0_CLK1P
MIPI_CSI0_CLK1N
MIPI_CSI0_D2P
MIPI_CSI0_D2N
MIPI_CSI0_D3P
MIPI_CSI0_D3N
MIPI_CSI0_AVCC0V75
MIPI_CSI0_AVCC1V8
AJ33 MIPI_DPHY2_RX_CLK0_P
AJ34 MIPI_DPHY2_RX_CLK0_N
AG33 MIPI_DPHY2_RX_D0_P
AG34 MIPI_DPHY2_RX_D0_N
AH33 MIPI_DPHY2_RX_D1_P
AH34 MIPI_DPHY2_RX_D1_N
AM33 MIPI_DPHY2_RX_CLK1_P
AM34 MIPI_DPHY2_RX_CLK1_N
AK33 MIPI_DPHY2_RX_D2_P
AK34 MIPI_DPHY2_RX_D2_N
AL33 MIPI_DPHY2_RX_D3_P
AL34 MIPI_DPHY2_RX_D3_N
MIPI_DPHY2_RX_CLK0_P
MIPI_DPHY2_RX_CLK0_N
MIPI_DPHY2_RX_D0_P
MIPI_DPHY2_RX_D0_N
MIPI_DPHY2_RX_D1_P
MIPI_DPHY2_RX_D1_N
MIPI_DPHY2_RX_CLK1_P
MIPI_DPHY2_RX_CLK1_N
MIPI_DPHY2_RX_D2_P
MIPI_DPHY2_RX_D2_N
MIPI_DPHY2_RX_D3_P
MIPI_DPHY2_RX_D3_N
VDDA_0V75
AB25
C101 0.1uF
C102 10uF
GND
AB26
VCCA1V8_IMAGE
C105 0.1uF
C106 10uF
GND
MIPI DPHY CSI_RX Port1
MIPI V1.2/2.5Gbps
MIPI_CSI1_CLK0P
MIPI_CSI1_CLK0N
MIPI_CSI1_D0P
MIPI_CSI1_D0N
MIPI_CSI1_D1P
MIPI_CSI1_D1N
MIPI_CSI1_CLK1P
MIPI_CSI1_CLK1N
MIPI_CSI1_D2P
MIPI_CSI1_D2N
MIPI_CSI1_D3P
MIPI_CSI1_D3N
MIPI_CSI1_AVCC0V75
MIPI_CSI1_AVCC1V8
AJ31 MIPI_DPHY3_RX_CLK0_P
AJ32 MIPI_DPHY3_RX_CLK0_N
AG31 MIPI_DPHY3_RX_D0_P
AG32 MIPI_DPHY3_RX_D0_N
AH31 MIPI_DPHY3_RX_D1_P
AH32 MIPI_DPHY3_RX_D1_N
AM31 MIPI_DPHY3_RX_CLK1_P
AM32 MIPI_DPHY3_RX_CLK1_N
AK31 MIPI_DPHY3_RX_D2_P
AK32 MIPI_DPHY3_RX_D2_N
AL31 MIPI_DPHY3_RX_D3_P
AL32 MIPI_DPHY3_RX_D3_N
MIPI_DPHY3_RX_CLK0_P
MIPI_DPHY3_RX_CLK0_N
MIPI_DPHY3_RX_D0_P
MIPI_DPHY3_RX_D0_N
MIPI_DPHY3_RX_D1_P
MIPI_DPHY3_RX_D1_N
MIPI_DPHY3_RX_CLK1_P
MIPI_DPHY3_RX_CLK1_N
MIPI_DPHY3_RX_D2_P
MIPI_DPHY3_RX_D2_N
MIPI_DPHY3_RX_D3_P
MIPI_DPHY3_RX_D3_N
VDDA_0V75
AA25
C115 0.1uF
C116 10uF
GND
AA26
VCCA1V8_IMAGE
C117 0.1uF
C118 10uF
GND
RK3588
U1U
SARADC
12-bit 1MS/s
Recovery/
SARADC_IN0_BOOT
SARADC_IN1
SARADC_IN2
SARADC_IN3
SARADC_IN4
SARADC_IN5
SARADC_IN6
SARADC_IN7
SARADC_AVDD_1V8
AM16 SARADC_IN0_BOOT
AL16 SARADC_IN1
AK16 SARADC_IN2
AN17 SARADC_IN3
AM17 SARADC_IN4
AK15 SARADC_IN5
AL17 SARADC_IN6
AK17 SARADC_IN7
SARADC_IN0_BOOT
SARADC_IN1
SARADC_IN2
SARADC_IN3
SARADC_IN4
SARADC_IN5
SARADC_IN6
SARADC_IN7
VCCA_1V8
AH18
C99 0.1uF
C100 10uF
GND
OTP
NC
AD4
OTP_VDDOTP_0V75
AD3
VDDA_0V75
C103 0.1uF
C104 10uF
GND
TSADC
TSADC_TEST_OUT_TS
AF18
RK3588
VCCA_1V8
R219 100K
R220 DNI/100K
GND
SARADC_IN0_BOOT C107 1nF
SARADC_IN1 C108 1nF
SARADC_IN2 C109 1nF
SARADC_IN3 C110 1nF
SARADC_IN4 C111 1nF
SARADC_IN5 C112 1nF
SARADC_IN6 C113 1nF
SARADC_IN7 C114 1nF
GND
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Number
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Date: 8-28-2026
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U1I

VCCIO4 Domain

Operating Voltage=1.8V/3.3V

			SPI4_MISO_M2	PCIE30X1_1_CLKREQN_M2	DP0_HPDIN_M2	I2C2_SDA_M4	UART6_RX_M1	GPIO1_A0_d
SATA1_ACT_LED_M1	/	/	SPI4_MOSI_M2	PCIE30X1_1_WAKEN_M2	DP1_HPDIN_M2	I2C2_SCL_M4	UART6_TX_M1	GPIO1_A1_d
		PWM0_M2	SPI4_CLK_M2	/	VOP_POST_EMPTY	I2C4_SDA_M3	UART6_RTSN_M1	GPIO1_A2_d
		PWM1_M2	SPI4_CS0_M2	/	HDMI_TX1_SDA_M2	I2C4_SCL_M3	UART6_CTSN_M1	GPIO1_A3_d
		/	SPI2_MISO_M0	/	HDMI_TX1_SCL_M2	/	/	GPIO1_A4_d
		/	SPI2_MOSI_M0		HDMI_TX0_HPD_M0	/	/	GPIO1_A5_d
		/	SPI2_CLK_M0	/	HDMI_TX1_HPD_M0	/	/	GPIO1_A6_d
		PWM3_IR_M3	SPI2_CS0_M0	PCIE30X1_1_PERSTN_M2	/	PDM1_SDI0_M1	/	GPIO1_A7_u
		/	SPI2_CS1_M0	PCIE30X4_CLKREQN_M3	/	PDM1_SDI1_M1	/	GPIO1_B0_u
		/	SPI0_MISO_M2	PCIE30X4_WAKEN_M3	/	PDM1_SDI2_M1	/	GPIO1_B1_d
		/	SPI0_MOSI_M2	PCIE30X4_PERSTN_M3	/	PDM1_SDI3_M1	UART4_RX_M2	GPIO1_B2_d
SATA0_ACT_LED_M1	/	/	SPI0_CLK_M2	PCIE30X1_0_WAKEN_M2	/	PDM1_CLK1_M1	UART4_TX_M2	GPIO1_B3_d
		/	SPI0_CS0_M2	PCIE30X1_0_PERSTN_M2	/	PDM1_CLK0_M1	UART7_RX_M2	GPIO1_B4_u
		/	SPI0_CS1_M2	PCIE30X1_0_CLKREQN_M2	/	/	UART7_TX_M2	GPIO1_B5_u
	HDMI_RX_HPDOUT_M2	/	SPDIF0_TX_M0	PCIE30X2_WAKEN_M3	MIPI_CAMERA1_CLK_M0	I2C5_SCL_M3	UART1_TX_M1	GPIO1_B6_u
SATA2_ACT_LED_M1	/ HDMI_RX_CEC_M2	/ PWM13_M2	SPDIF1_TX_M0	PCIE30X2_PERSTN_M3	MIPI_CAMERA2_CLK_M0	I2C5_SDA_M3	UART1_RX_M1	GPIO1_B7_u
	HDMI_RX_SCL_M2	/ PWM14_M2	/	/	MIPI_CAMERA3_CLK_M0	I2C8_SCL_M2	UART1_RTSN_M1	GPIO1_D6_u
	HDMI_RX_SDA_M2	/ PWM15_IR_M3	/	PCIE30X2_CLKREQN_M3	MIPI_CAMERA4_CLK_M0	I2C8_SDA_M2	UART1_CTSN_M1	GPIO1_D7_u

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A24 B4_GPIO1_A0 B4_GPIO1_A0

A25 B4_GPIO1_A1 B4_GPIO1_A1

A26 B4_GPIO1_A2 B4_GPIO1_A2

A27 B4_GPIO1_A3 B4_GPIO1_A3

B25 B4_GPIO1_A4 B4_GPIO1_A4

B26 B4_GPIO1_A5 B4_GPIO1_A5

C24 B4_GPIO1_A6 B4_GPIO1_A6

C25 B4_GPIO1_A7 B4_GPIO1_A7

C27 B4_GPIO1_B0 B4_GPIO1_B0

D25 B4_GPIO1_B1 B4_GPIO1_B1

D26 B4_GPIO1_B2 B4_GPIO1_B2

D27 B4_GPIO1_B3 B4_GPIO1_B3

E24 B4_GPIO1_B4 B4_GPIO1_B4

E25 B4_GPIO1_B5 B4_GPIO1_B5

E26 B4_GPIO1_B6 B4_GPIO1_B6

E27 B4_GPIO1_B7 B4_GPIO1_B7

F24 B4_GPIO1_D6 B4_GPIO1_D6

F25 B4_GPIO1_D7 B4_GPIO1_D7

H20 VCCIO4_1V8

VCC_1V8

C119 0.1uF

C120 10uF

GND

H21 VCCIO4

RK_VCCIO4

C121 0.1uF

C122 10uF

GND

FB1 BLM18KG121TN1D

VCC_1V8

FB2 DNI/BLM18KG121TN1D

VCC_3V3

Title		
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U1K

VCCIO6 Domain

Operating Voltage=1.8V/3.3V

	SPI0_MISO_M1	UART9_RTSN_M1		I2S1_MCLK_M0	PCIE30X1_1_CLKREQN_M1	BT1120_D0	CIF_D0	GPIO4_A0_d
	SPI0_MOSI_M1	UART9_CTSN_M1		I2S1_SCLK_M0	PCIE30X1_1_WAKEN_M1	BT1120_D1	CIF_D1	GPIO4_A1_d
	SPI0_CLK_M1			I2S1_LRCK_M0	PCIE30X1_1_PERSTN_M1	BT1120_D2	CIF_D2	GPIO4_A2_d
		UART0_TX_M2			PCIE30X1_0_CLKREQN_M1	BT1120_D3	CIF_D3	GPIO4_A3_d
	SPI2_MISO_M1	UART0_RX_M2	I2C3_SCL_M2		PCIE30X1_0_WAKEN_M1	BT1120_D4	CIF_D4	GPIO4_A4_d
	SPI2_MOSI_M1	UART3_TX_M2	I2C3_SDA_M2	I2S1_SDI0_M0	PCIE30X1_0_PERSTN_M1	BT1120_D5	CIF_D5	GPIO4_A5_d
	SPI2_CLK_M1	UART3_RX_M2	I2C5_SCL_M2	I2S1_SDI1_M0	PCIE30X2_CLKREQN_M1	BT1120_D6	CIF_D6	GPIO4_A6_d
	SPI2_CS0_M1		I2C5_SDA_M2	I2S1_SDI2_M0	PCIE30X2_WAKEN_M1	BT1120_D7	CIF_D7	GPIO4_A7_d
	SPI2_CS1_M1	UART8_TX_M0	I2C6_SDA_M3	I2S1_SDI3_M0	PCIE30X2_PERSTN_M1	BT1120_CLKOUT	CIF_CLKIN	GPIO4_B0_d
SATA2_ACT_LED_M0	SPDIF1_TX_M1	SPI0_CS1_M1	UART8_RX_M0	I2C6_SCL_M3	I2S1_SDO0_M0	PCIE30X1_0_BUTTON_RSTN		MIPI_CAMERA0_CLK_M0	GPIO4_B1_u
CAN1_RX_M1	PWM14_M1	SPI0_CS0_M1	UART8_RTSN_M0	I2C7_SCL_M3	I2S1_SDO1_M0	PCIE30X1_1_BUTTON_RSTN	BT1120_D8	CIF_HREF	GPIO4_B2_u
CAN1_TX_M1	PWM15_IR_M1		UART8_CTSN_M0	I2C7_SDA_M3	I2S1_SDO2_M0	PCIE20X1_2_BUTTON_RSTN	BT1120_D9	CIF_VSYNC	GPIO4_B3_u
SPDIF0_TX_M1	PWM11_IR_M1	DP0_HPDIN_M0	UART9_TX_M1		I2S1_SDO3_M0	PCIE30X4_CLKREQN_M1	BT1120_D10	CIF_CLKOUT	GPIO4_B4_u
SATA1_ACT_LED_M0	PWM12_M1	SPI3_MISO_M1	UART9_RX_M1		HDMI_RX_CEC_M0	PCIE30X4_WAKEN_M1	BT1120_D11		GPIO4_B5_d
SATA0_ACT_LED_M0	PWM13_M1	SPI3_MOSI_M1		I2C5_SCL_M1	HDMI_RX_HPDOUT_M0	PCIE30X4_PERSTN_M1	BT1120_D12		GPIO4_B6_d
		SPI3_CLK_M1		I2C5_SDA_M1	HDMI_TX0_SCL_M0	PCIE20X1_2_CLKREQN_M1	BT1120_D13		GPIO4_B7_u
		SPI3_CS0_M1		I2C8_SCL_M3	HDMI_TX0_SDA_M0	PCIE20X1_2_WAKEN_M1	BT1120_D14		GPIO4_C0_u
SPDIF1_TX_M2	PWM6_M1	SPI3_CS1_M1		I2C8_SDA_M3	HDMI_TX0_CEC_M0	PCIE20X1_2_PERSTN_M1	BT1120_D15		GPIO4_C1_d

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AK30	B6_GPIO4_A0	B6_GPIO4_A0
AL30	B6_GPIO4_A1	B6_GPIO4_A1
AM29	B6_GPIO4_A2	B6_GPIO4_A2
AL29	B6_GPIO4_A3	B6_GPIO4_A3
AL28	B6_GPIO4_A4	B6_GPIO4_A4
AK27	B6_GPIO4_A5	B6_GPIO4_A5
AL27	B6_GPIO4_A6	B6_GPIO4_A6
AM27	B6_GPIO4_A7	B6_GPIO4_A7
AK26	B6_GPIO4_B0	B6_GPIO4_B0
AL24	B6_GPIO4_B1	B6_GPIO4_B1
AK25	B6_GPIO4_B2	B6_GPIO4_B2
AM25	B6_GPIO4_B3	B6_GPIO4_B3
AL26	B6_GPIO4_B4	B6_GPIO4_B4
AJ26	B6_GPIO4_B5	B6_GPIO4_B5
AJ27	B6_GPIO4_B6	B6_GPIO4_B6
AJ28	B6_GPIO4_B7	B6_GPIO4_B7
AJ25	B6_GPIO4_C0	B6_GPIO4_C0
AK24	B6_GPIO4_C1	B6_GPIO4_C1

VCCIO6_1V8 — AC25

VCC_1V8

C123 0.1uF

C124 10uF

GND

VCCIO6 — AC26

RK_VCCIO6

C125 0.1uF

C126 10uF

GND

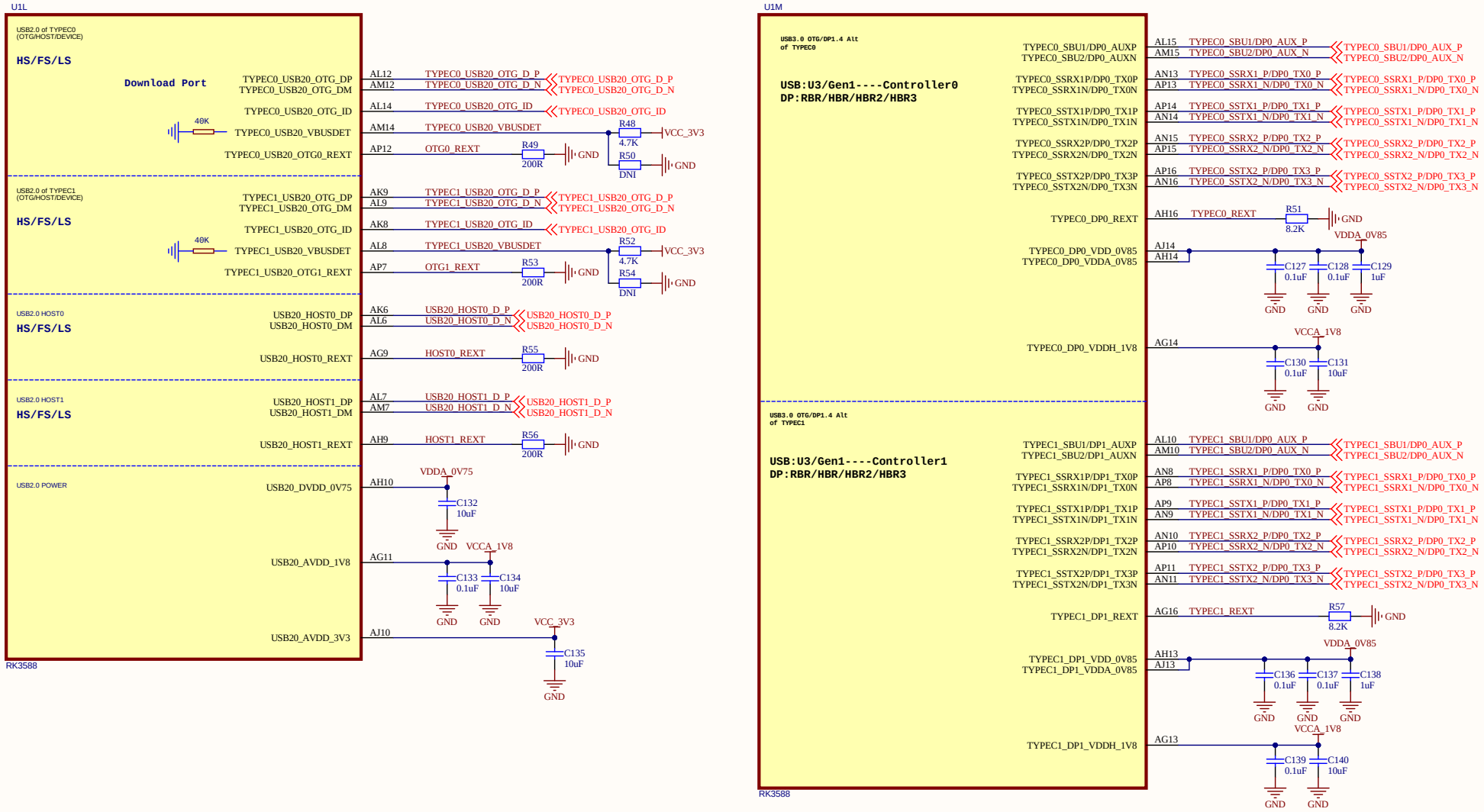
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VCC_1V8

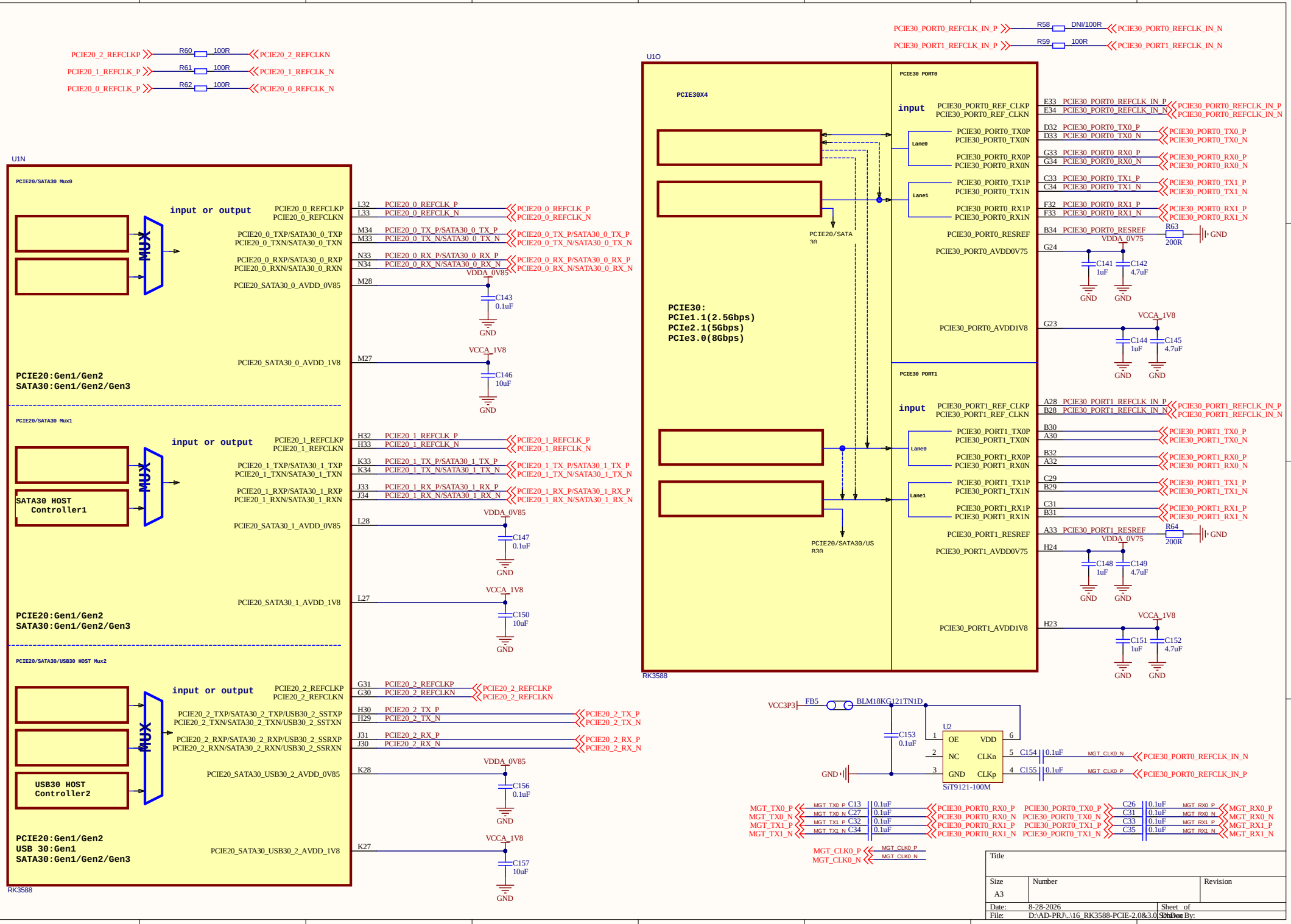
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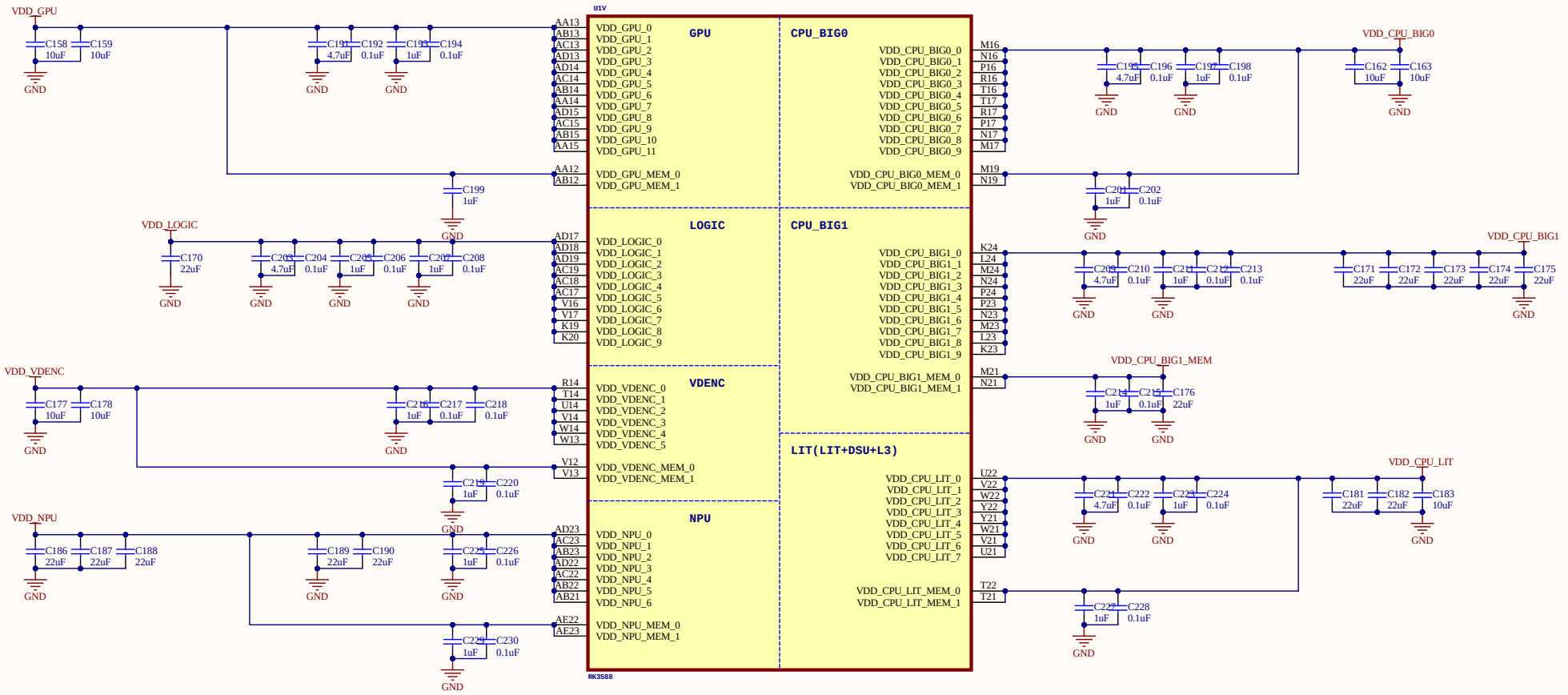
VCC_3V3

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PCIE20_2_REFCLKP
R60
100R
PCIE20_2_REFCLKN
PCIE20_1_REFCLK_P
R61
100R
PCIE20_1_REFCLK_N
PCIE20_0_REFCLK_P
R62
100R
PCIE20_0_REFCLK_N
U1N
PCIE20/SATA30 Mux0
input or output
PCIE20_0_REFCLKP
PCIE20_0_REFCLKN
PCIE20_0_TXP/SATA30_0_TXP
PCIE20_0_TXN/SATA30_0_TXN
PCIE20_0_RXP/SATA30_0_RXP
PCIE20_0_RXN/SATA30_0_RXN
PCIE20_SATA30_0_AVDD_0V85
PCIE20_SATA30_0_AVDD_1V8
PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3
L32 PCIE20_0_REFCLK_P
L33 PCIE20_0_REFCLK_N
M34 PCIE20_0_TX_P/SATA30_0_TX_P
M33 PCIE20_0_TX_N/SATA30_0_TX_N
N33 PCIE20_0_RX_P/SATA30_0_RX_P
N34 PCIE20_0_RX_N/SATA30_0_RX_N
M28
M27
VDDA_0V85
VCCA_1V8
C143 0.1uF
C146 10uF
GND
PCIE20/SATA30 Mux1
SATA30 HOST Controller1
H32 PCIE20_1_REFCLK_P
H33 PCIE20_1_REFCLK_N
K33 PCIE20_1_TX_P/SATA30_1_TX_P
K34 PCIE20_1_TX_N/SATA30_1_TX_N
J33 PCIE20_1_RX_P/SATA30_1_RX_P
J34 PCIE20_1_RX_N/SATA30_1_RX_N
L28
L27
C147 0.1uF
C150 10uF
PCIE20/SATA30/USB30 HOST Mux2
USB30 HOST Controller2
PCIE20:Gen1/Gen2
USB 30:Gen1
SATA30:Gen1/Gen2/Gen3
G31 PCIE20_2_REFCLKP
G30 PCIE20_2_REFCLKN
H30 PCIE20_2_TX_P
H29 PCIE20_2_TX_N
J31 PCIE20_2_RX_P
J30 PCIE20_2_RX_N
K28
K27
C156 0.1uF
C157 10uF
RK3588
PCIE30_PORT0_REFCLK_IN_P
R58
DNI/100R
PCIE30_PORT0_REFCLK_IN_N
PCIE30_PORT1_REFCLK_IN_P
R59
100R
PCIE30_PORT1_REFCLK_IN_N
U1O
PCIE30X4
PCIE30 PORT0
PCIE30 PORT1
input
PCIE30_PORT0_REF_CLKP
PCIE30_PORT0_REF_CLKN
PCIE30_PORT1_REF_CLKP
PCIE30_PORT1_REF_CLKN
Lane0
Lane1
PCIE30_PORT0_TX0P
PCIE30_PORT0_TX0N
PCIE30_PORT0_RX0P
PCIE30_PORT0_RX0N
PCIE30_PORT0_TX1P
PCIE30_PORT0_TX1N
PCIE30_PORT0_RX1P
PCIE30_PORT0_RX1N
PCIE30_PORT0_RESREF
PCIE30_PORT0_AVDD0V75
PCIE30_PORT0_AVDD1V8
PCIE30_PORT1_TX0P
PCIE30_PORT1_TX0N
PCIE30_PORT1_RX0P
PCIE30_PORT1_RX0N
PCIE30_PORT1_TX1P
PCIE30_PORT1_TX1N
PCIE30_PORT1_RX1P
PCIE30_PORT1_RX1N
PCIE30_PORT1_RESREF
PCIE30_PORT1_AVDD0V75
PCIE30_PORT1_AVDD1V8
PCIE20/SATA 3G
PCIE20/SATA30/US B3G
PCIE30:
PCIe1.1(2.5Gbps)
PCIe2.1(5Gbps)
PCIe3.0(8Gbps)
RK3588
E33 PCIE30_PORT0_REFCLK_IN_P
E34 PCIE30_PORT0_REFCLK_IN_N
D32 PCIE30_PORT0_TX0_P
D33 PCIE30_PORT0_TX0_N
G33 PCIE30_PORT0_RX0_P
G34 PCIE30_PORT0_RX0_N
C33 PCIE30_PORT0_TX1_P
C34 PCIE30_PORT0_TX1_N
F32 PCIE30_PORT0_RX1_P
F33 PCIE30_PORT0_RX1_N
B34 PCIE30_PORT0_RESREF
R63 200R
G24
G23
C141 1uF
C142 4.7uF
C144 1uF
C145 4.7uF
A28 PCIE30_PORT1_REFCLK_IN_P
B28 PCIE30_PORT1_REFCLK_IN_N
B30
A30
B32
A32
C29
B29
C31
B31
A33 PCIE30_PORT1_RESREF
R64 200R
H24
H23
C148 1uF
C149 4.7uF
C151 1uF
C152 4.7uF
VDDA_0V75
VCCA_1V8
GND
VCC3P3
FB5
BLM18KG121TN1D
C153 0.1uF
U2
OE
VDD
NC
CLKn
GND
CLKp
SiT9121-100M
C154 0.1uF
C155 0.1uF
MGT_CLK0_N
MGT_CLK0_P
MGT_TX0_P
MGT_TX0_N
MGT_TX1_P
MGT_TX1_N
C13 0.1uF
C27 0.1uF
C32 0.1uF
C34 0.1uF
C26 0.1uF
C31 0.1uF
C33 0.1uF
C35 0.1uF
MGT_RX0_P
MGT_RX0_N
MGT_RX1_P
MGT_RX1_N
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Size
A3
Number
Revision
Date:
8-28-2026
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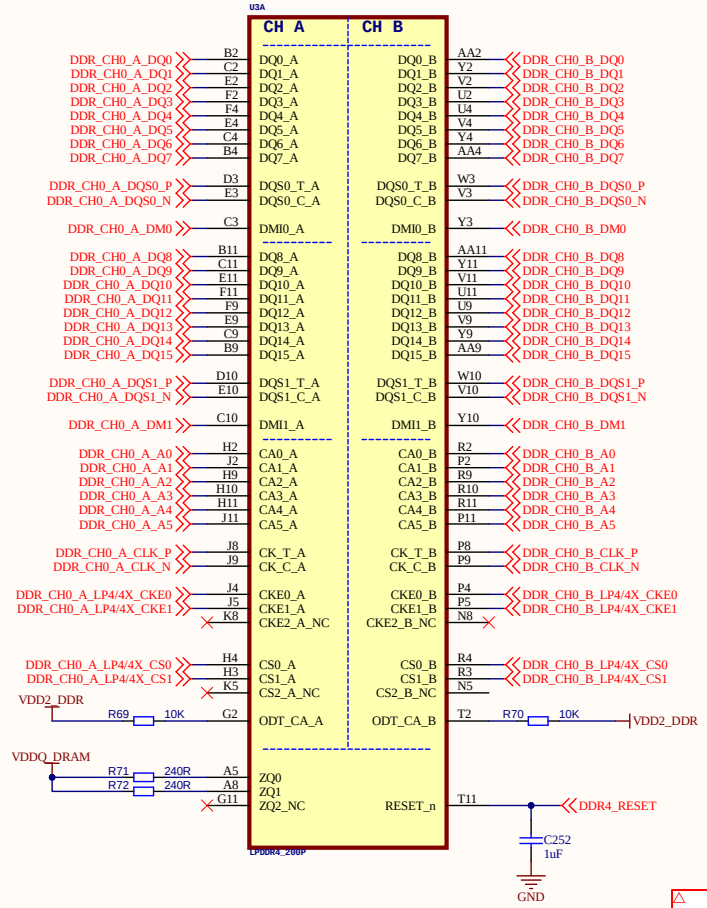


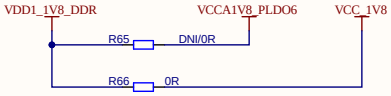
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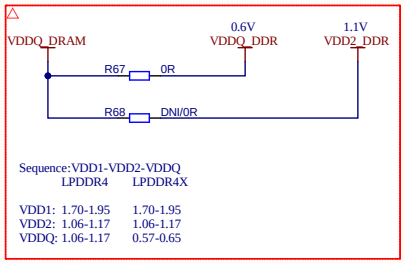


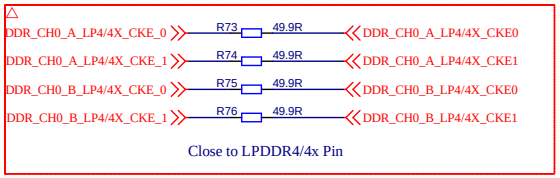
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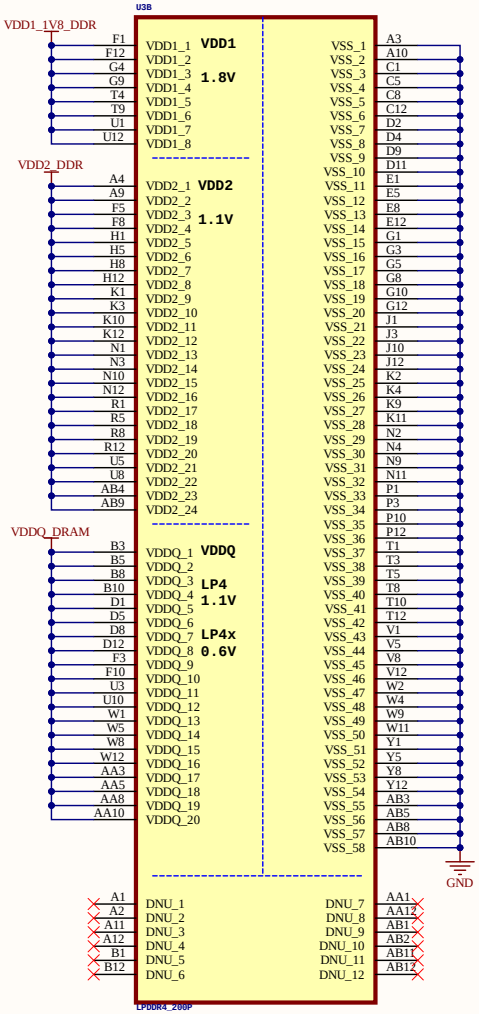
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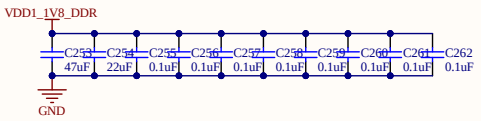


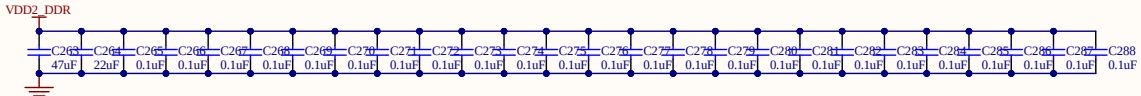


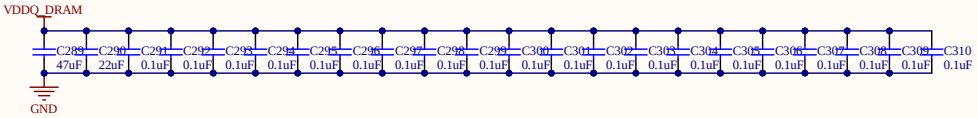






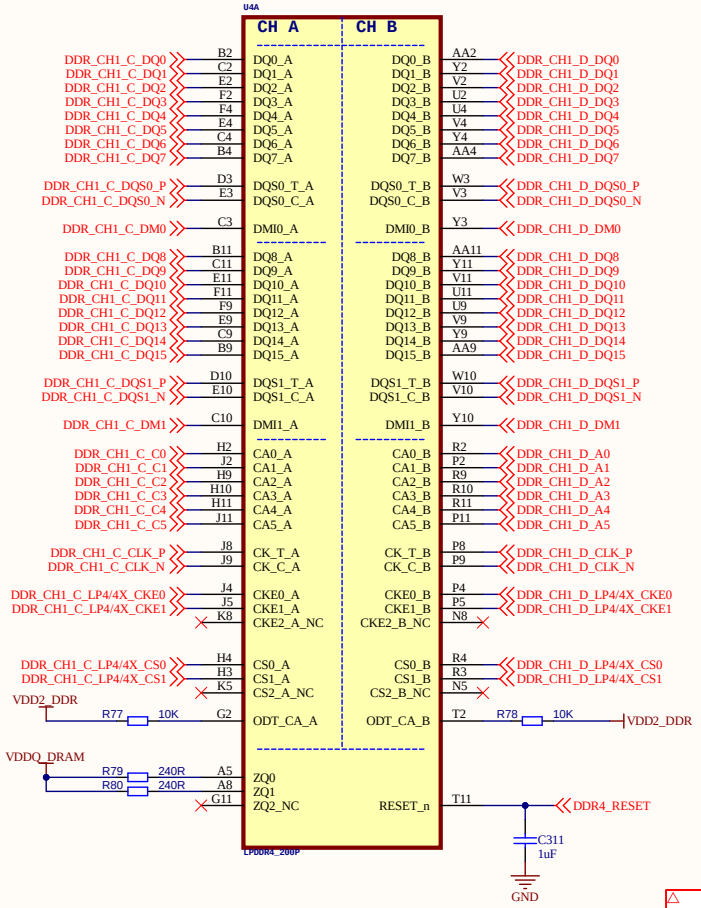


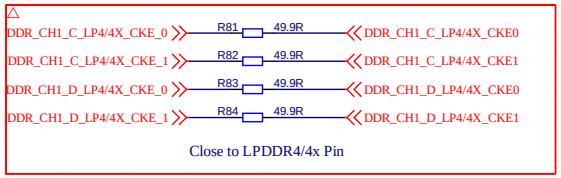


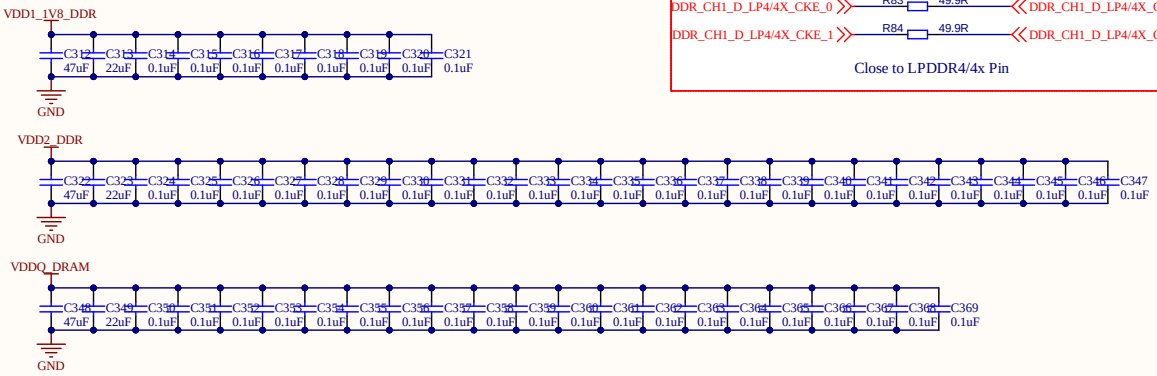


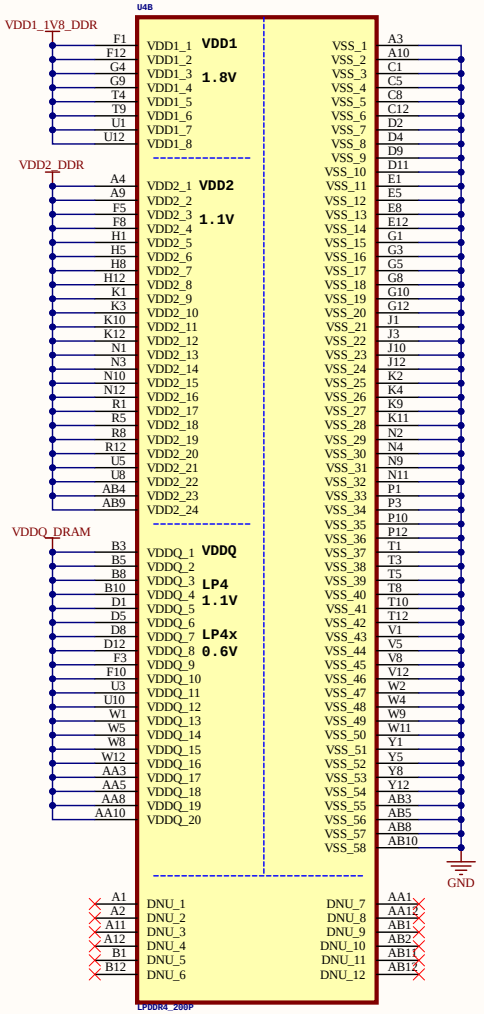
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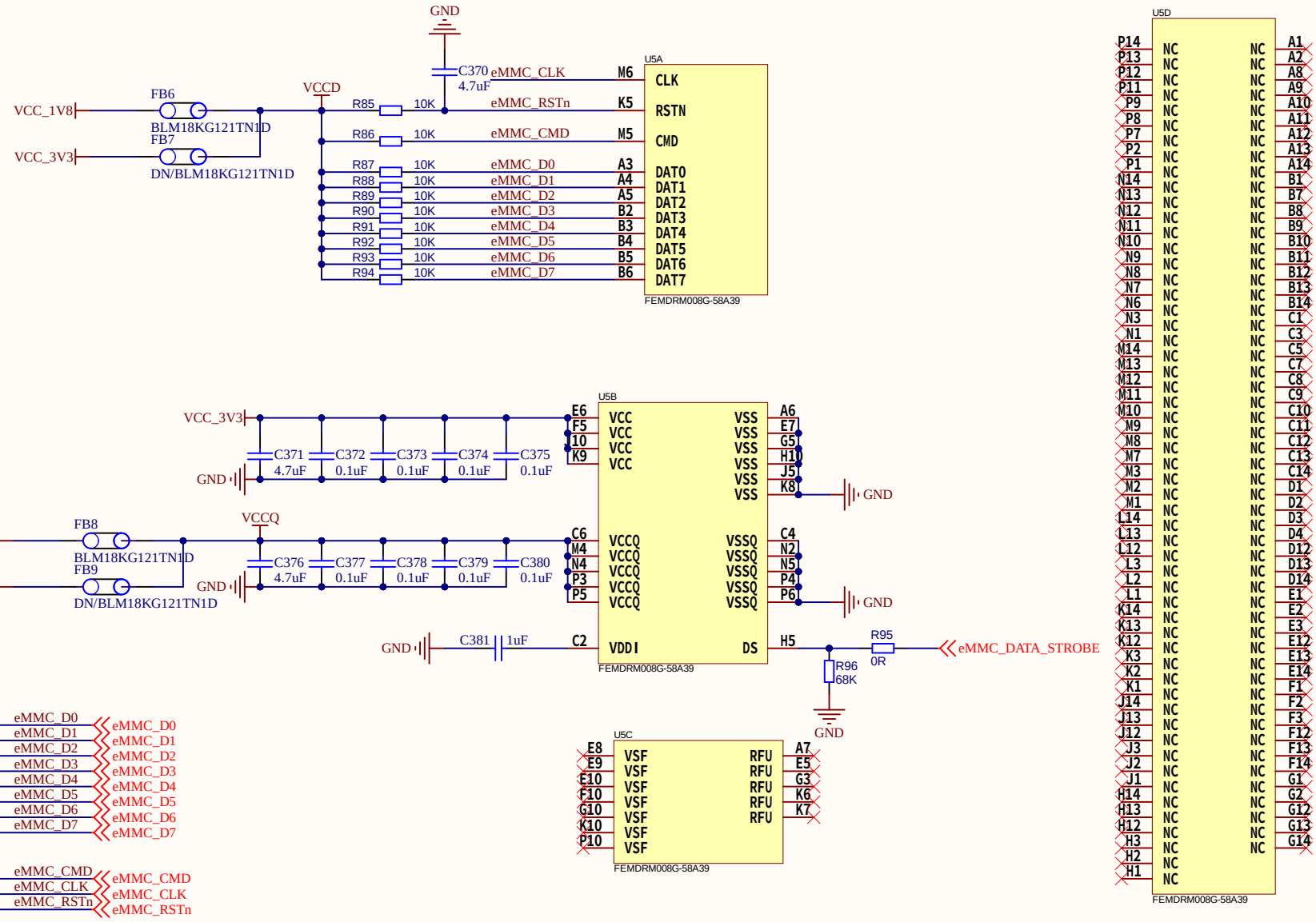


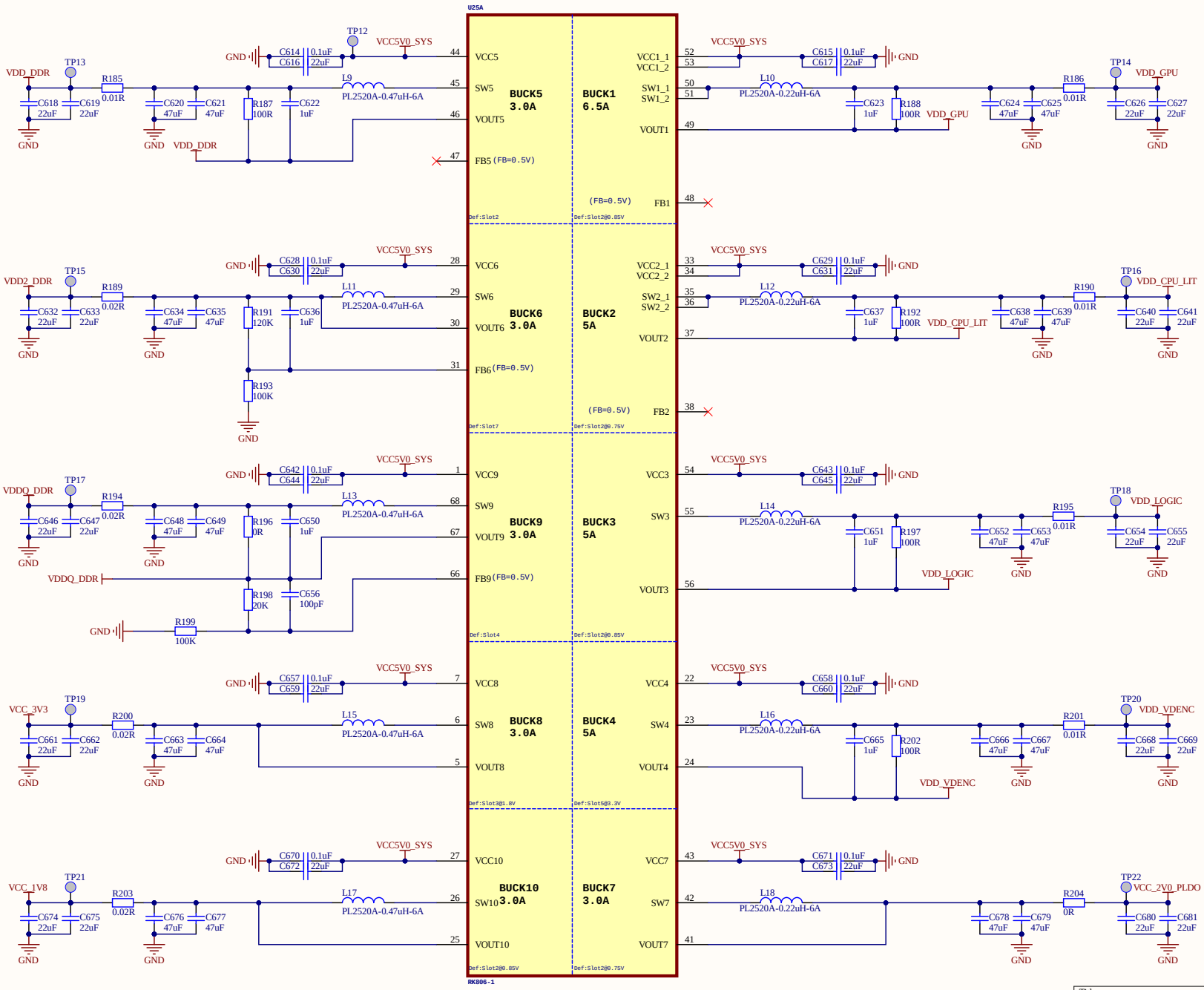




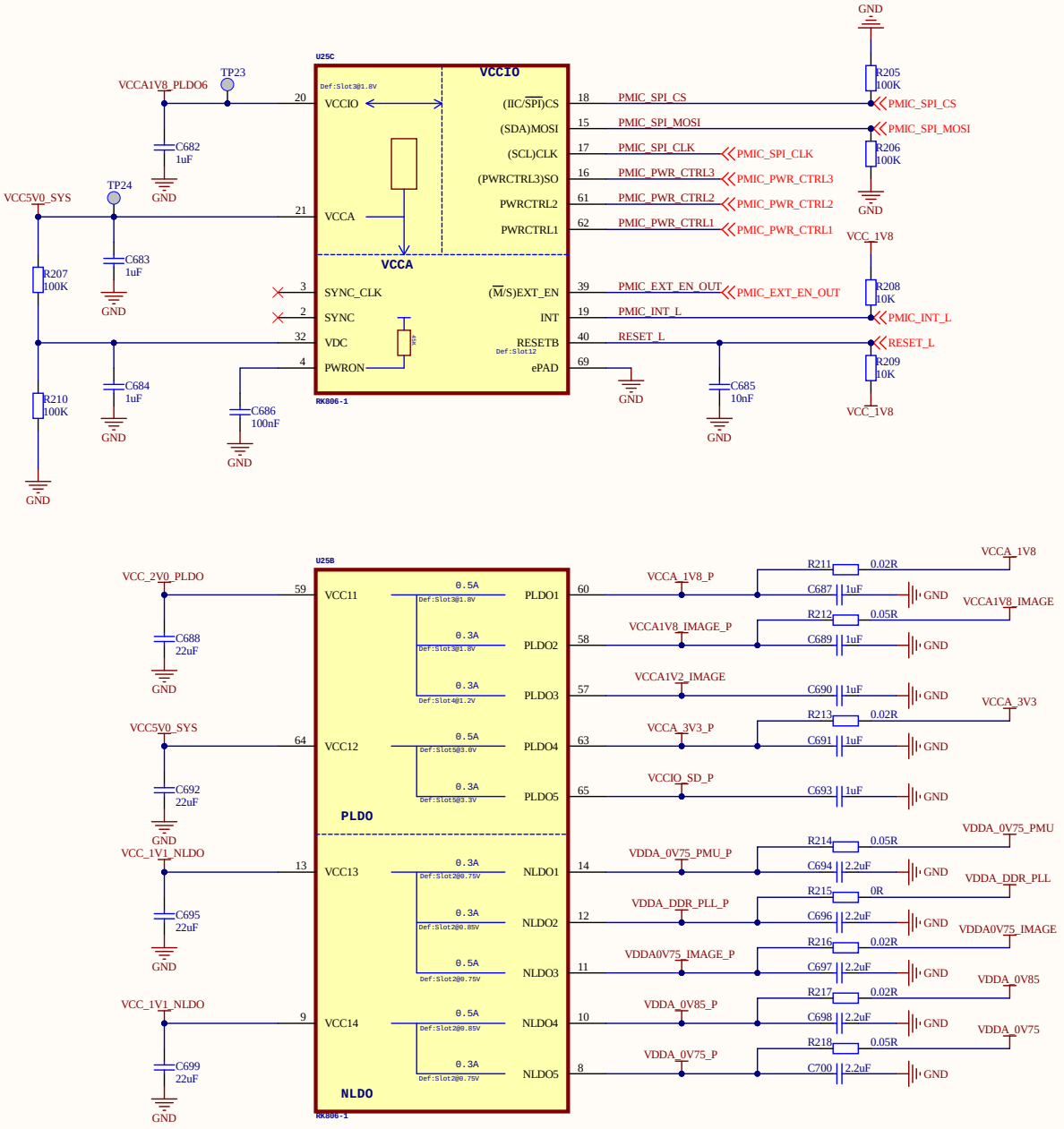


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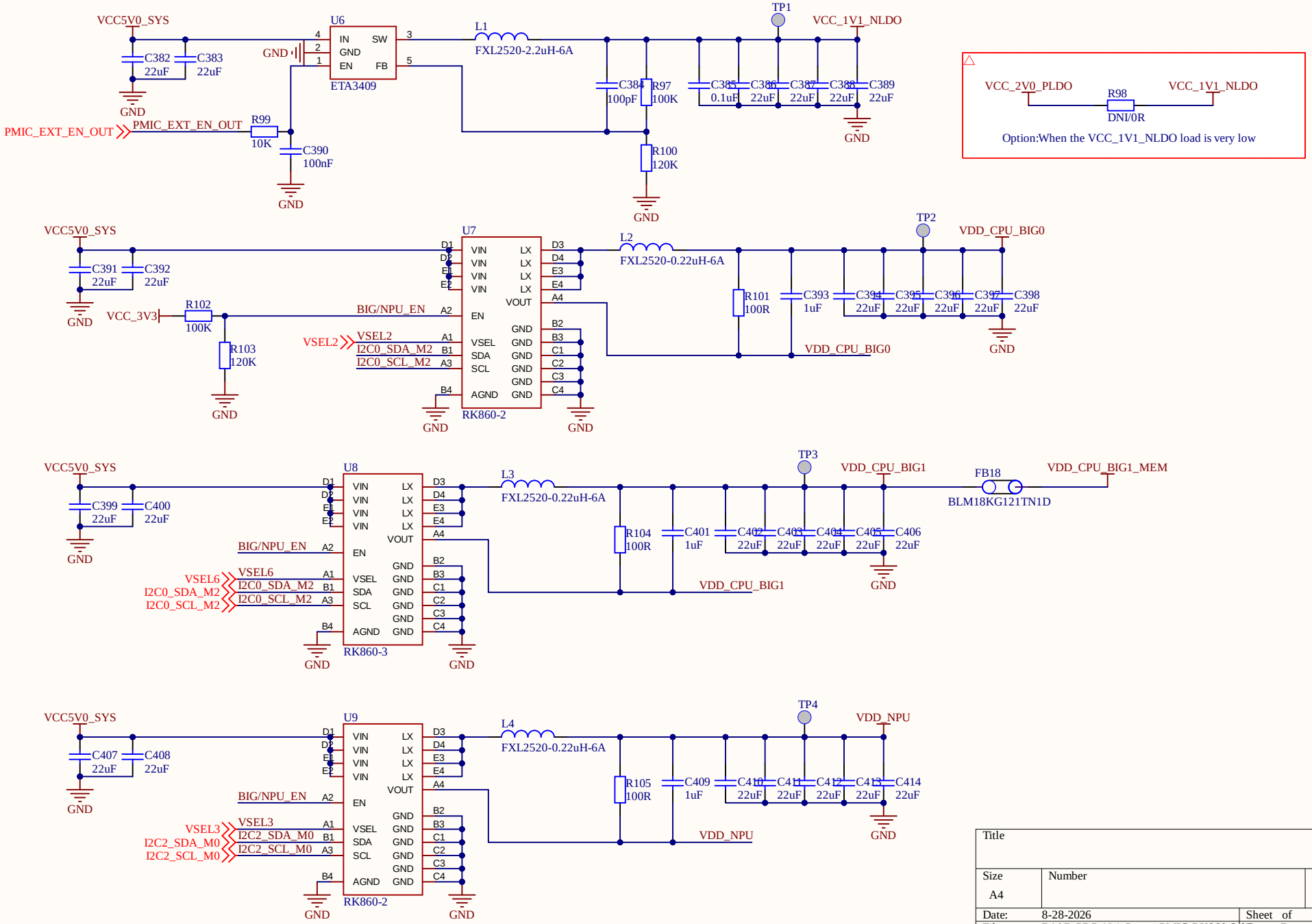

GND
C370
4.7uF
eMMC_CLK
eMMC_RSTn
eMMC_CMD
eMMC_D0
eMMC_D1
eMMC_D2
eMMC_D3
eMMC_D4
eMMC_D5
eMMC_D6
eMMC_D7
VCCD
FB6
VCC_1V8
BLM18KG121TN1D
FB7
VCC_3V3
DN/BLM18KG121TN1D
R85 10K
R86 10K
R87 10K
R88 10K
R89 10K
R90 10K
R91 10K
R92 10K
R93 10K
R94 10K
U5A
M6 CLK
K5 RSTN
M5 CMD
A3 DAT0
A4 DAT1
A5 DAT2
B2 DAT3
B3 DAT4
B4 DAT5
B5 DAT6
B6 DAT7
FEMDRM008G-58A39
U5B
VCC_3V3
C371 4.7uF
C372 0.1uF
C373 0.1uF
C374 0.1uF
C375 0.1uF
E6 VCC
F5 VCC
J10 VCC
K9 VCC
A6 VSS
E7 VSS
G5 VSS
H10 VSS
J5 VSS
K8 VSS
VSS
VCCQ
FB8
VCC_1V8
BLM18KG121TN1D
FB9
VCC_3V3
DN/BLM18KG121TN1D
C376 4.7uF
C377 0.1uF
C378 0.1uF
C379 0.1uF
C380 0.1uF
C6 VCCQ
M4 VCCQ
N4 VCCQ
P3 VCCQ
P5 VCCQ
C4 VSSQ
N2 VSSQ
N5 VSSQ
P4 VSSQ
P6 VSSQ
C381 1uF
C2 VDDI
H5 DS
R95 0R
R96 68K
eMMC_DATA_STROBE
U5C
E8 VSF
E9 VSF
E10 VSF
F10 VSF
G10 VSF
K10 VSF
P10 VSF
A7 RFU
E5 RFU
G3 RFU
K6 RFU
K7 RFU
FEMDRM008G-58A39
U5D
P14 NC
P13 NC
P12 NC
P11 NC
P9 NC
P8 NC
P7 NC
P2 NC
P1 NC
N14 NC
N13 NC
N12 NC
N11 NC
N10 NC
N9 NC
N8 NC
N7 NC
N6 NC
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N1 NC
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M13 NC
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M8 NC
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M1 NC
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K14 NC
K13 NC
K12 NC
K3 NC
K2 NC
K1 NC
J14 NC
J13 NC
J12 NC
J3 NC
J2 NC
J1 NC
H14 NC
H13 NC
H12 NC
H3 NC
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H1 NC
A1 NC
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A11 NC
A12 NC
A13 NC
A14 NC
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B13 NC
B14 NC
C1 NC
C3 NC
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C7 NC
C8 NC
C9 NC
C10 NC
C11 NC
C12 NC
C13 NC
C14 NC
D1 NC
D2 NC
D3 NC
D4 NC
D12 NC
D13 NC
D14 NC
E1 NC
E2 NC
E3 NC
E12 NC
E13 NC
E14 NC
F1 NC
F2 NC
F3 NC
F12 NC
F13 NC
F14 NC
G1 NC
G2 NC
G12 NC
G13 NC
G14 NC
FEMDRM008G-58A39
eMMC_D0
eMMC_D1
eMMC_D2
eMMC_D3
eMMC_D4
eMMC_D5
eMMC_D6
eMMC_D7
eMMC_CMD
eMMC_CLK
eMMC_RSTn
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Revision
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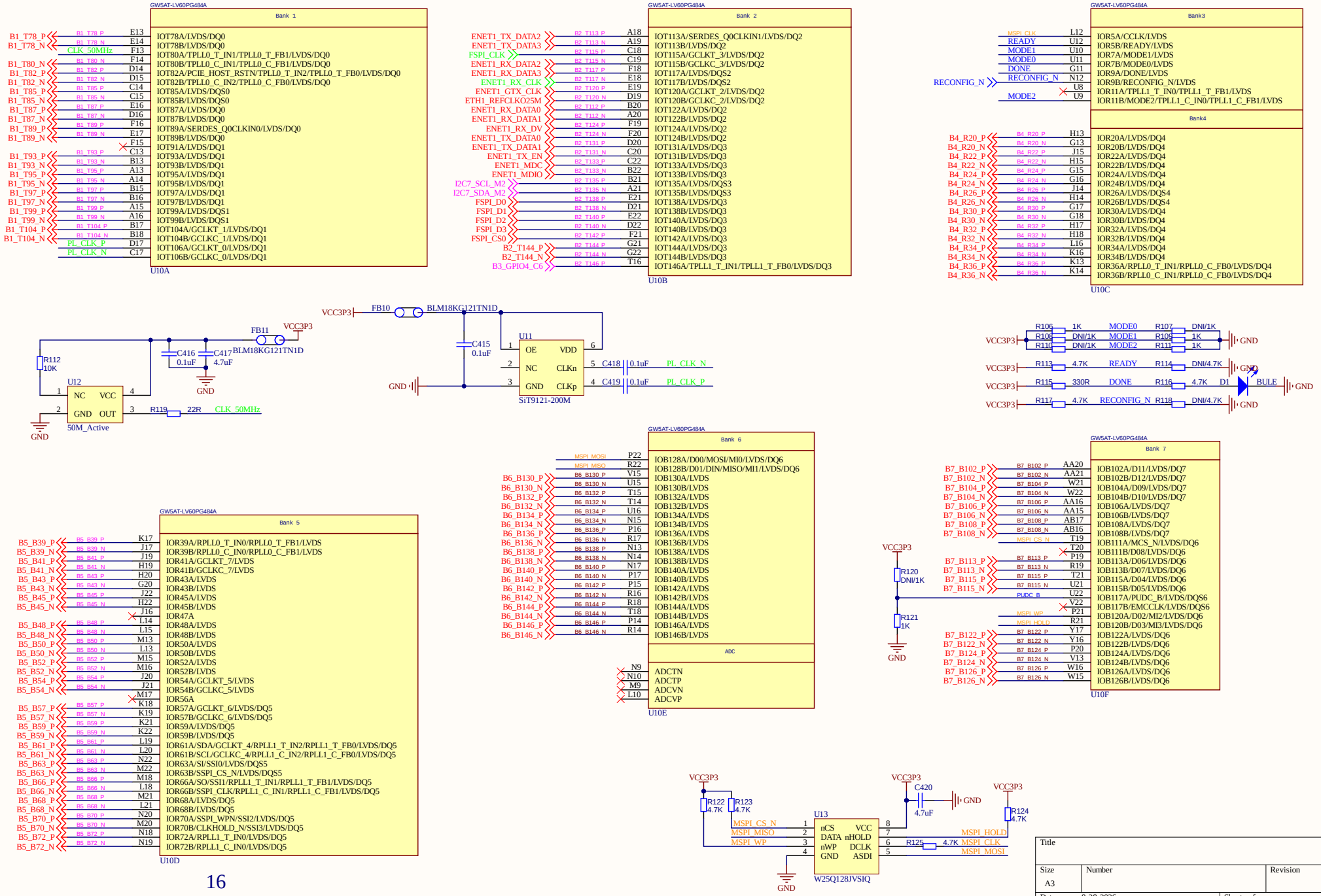


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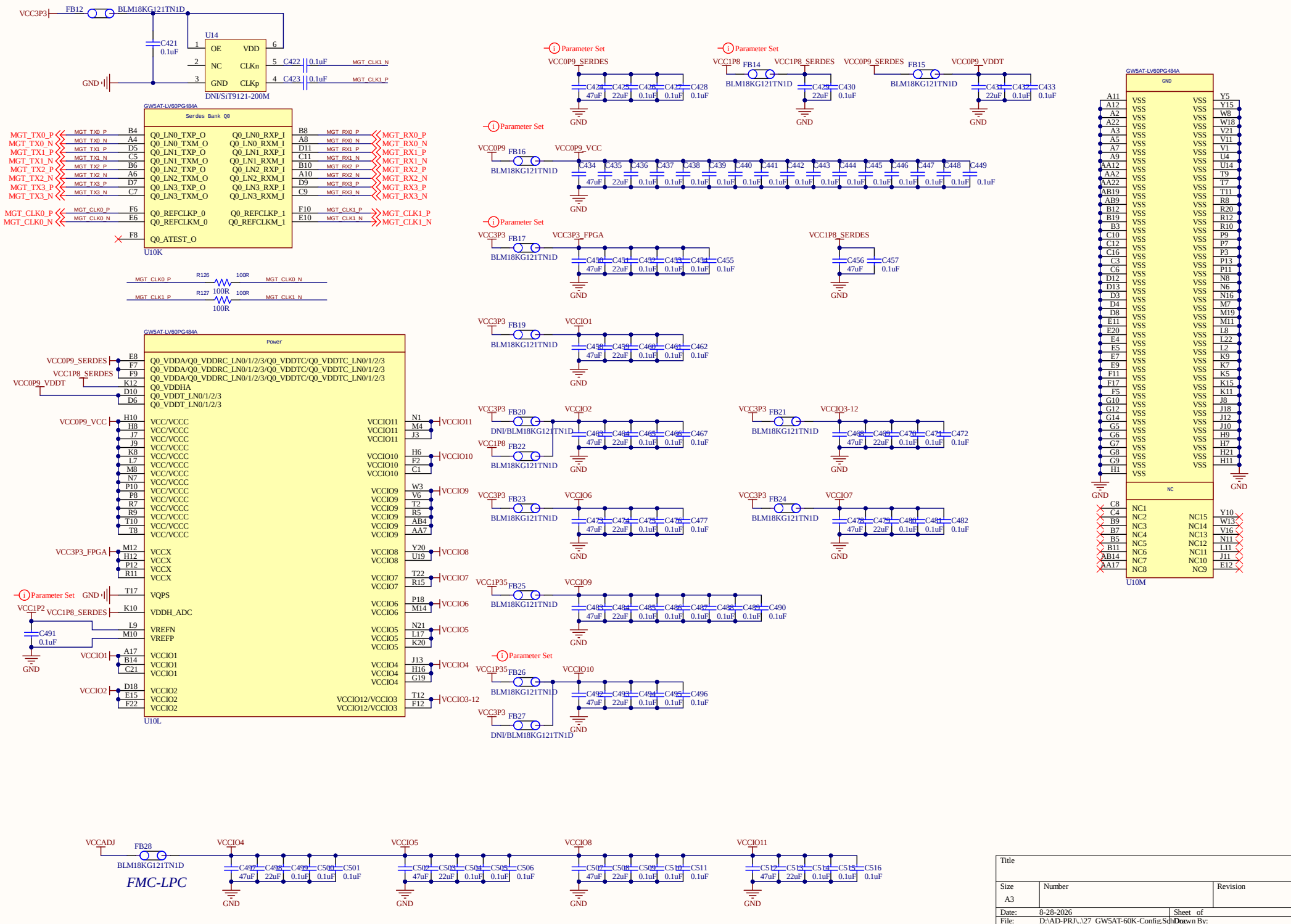


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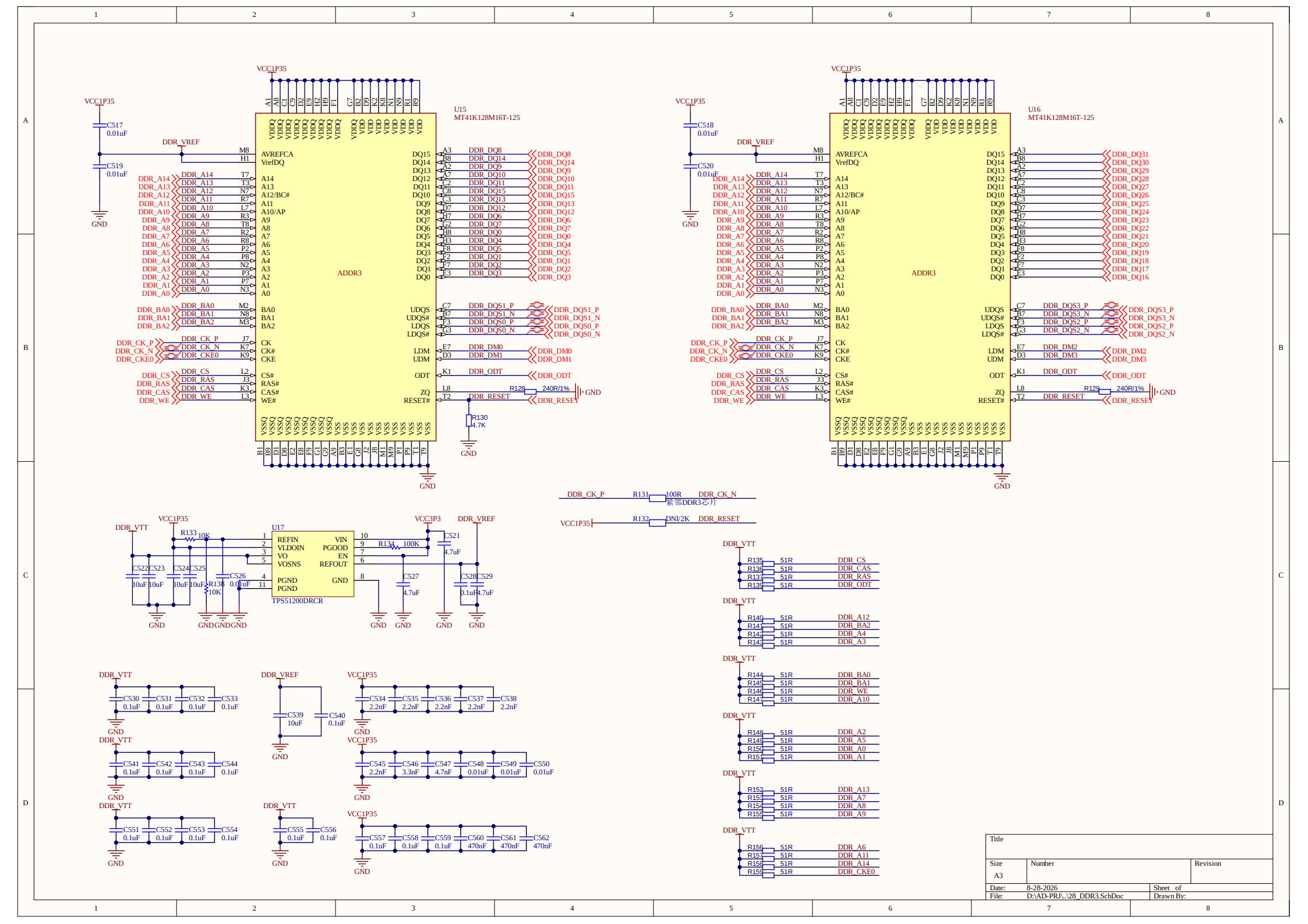

VCC5V0_SYS
U6
IN
SW
GND
EN
FB
ETA3409
C382
22uF
C383
22uF
GND
PMIC_EXT_EN_OUT
R99
10K
C390
100nF
L1
FXL2520-2.2uH-6A
TP1
VCC_1V1_NLDO
C384
100pF
R97
100K
C385
0.1uF
C386
22uF
C387
22uF
C388
22uF
C389
22uF
R100
120K
VCC_2V0_PLDO
R98
DNI/0R
Option:When the VCC_1V1_NLDO load is very low
U7
VIN
LX
VOUT
EN
VSEL
SDA
SCL
AGND
RK860-2
C391
22uF
C392
22uF
VCC_3V3
R102
100K
R103
120K
BIG/NPU_EN
VSEL2
I2C0_SDA_M2
I2C0_SCL_M2
L2
FXL2520-0.22uH-6A
TP2
VDD_CPU_BIG0
R101
100R
C393
1uF
C394
22uF
C395
22uF
C396
22uF
C397
22uF
C398
22uF
U8
RK860-3
C399
22uF
C400
22uF
VSEL6
L3
FXL2520-0.22uH-6A
TP3
VDD_CPU_BIG1
FB18
BLM18KG121TN1D
VDD_CPU_BIG1_MEM
R104
100R
C401
1uF
C402
22uF
C403
22uF
C404
22uF
C405
22uF
C406
22uF
U9
RK860-2
C407
22uF
C408
22uF
VSEL3
I2C2_SDA_M0
I2C2_SCL_M0
L4
FXL2520-0.22uH-6A
TP4
VDD_NPU
R105
100R
C409
1uF
C410
22uF
C411
22uF
C412
22uF
C413
22uF
C414
22uF
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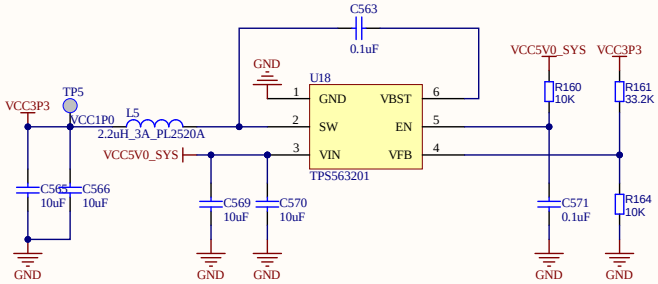

GW5AT-LV60PG484A
Bank 1
IOT78A/LVDS/DQ0
IOT78B/LVDS/DQ0
IOT80A/TPLL0_T_IN1/TPLL0_T_FB1/LVDS/DQ0
IOT80B/TPLL0_C_IN1/TPLL0_C_FB1/LVDS/DQ0
IOT82A/PCIE_HOST_RSTN/TPLL0_T_IN2/TPLL0_T_FB0/LVDS/DQ0
IOT82B/TPLL0_C_IN2/TPLL0_C_FB0/LVDS/DQ0
IOT85A/LVDS/DQS0
IOT85B/LVDS/DQS0
IOT87A/LVDS/DQ0
IOT87B/LVDS/DQ0
IOT89A/SERDES_Q0CLKIN0/LVDS/DQ0
IOT89B/LVDS/DQ0
IOT91A/LVDS/DQ1
IOT93A/LVDS/DQ1
IOT93B/LVDS/DQ1
IOT95A/LVDS/DQ1
IOT95B/LVDS/DQ1
IOT97A/LVDS/DQ1
IOT97B/LVDS/DQ1
IOT99A/LVDS/DQS1
IOT99B/LVDS/DQS1
IOT104A/GCLKT_1/LVDS/DQ1
IOT104B/GCLKC_1/LVDS/DQ1
IOT106A/GCLKT_0/LVDS/DQ1
IOT106B/GCLKC_0/LVDS/DQ1
U10A
GW5AT-LV60PG484A
Bank 2
IOT113A/SERDES_Q0CLKIN1/LVDS/DQ2
IOT113B/LVDS/DQ2
IOT115A/GCLKT_3/LVDS/DQ2
IOT115B/GCLKC_3/LVDS/DQ2
IOT117A/LVDS/DQS2
IOT117B/LVDS/DQS2
IOT120A/GCLKT_2/LVDS/DQ2
IOT120B/GCLKC_2/LVDS/DQ2
IOT122A/LVDS/DQ2
IOT122B/LVDS/DQ2
IOT124A/LVDS/DQ2
IOT124B/LVDS/DQ2
IOT131A/LVDS/DQ3
IOT131B/LVDS/DQ3
IOT133A/LVDS/DQ3
IOT133B/LVDS/DQ3
IOT135A/LVDS/DQS3
IOT135B/LVDS/DQS3
IOT138A/LVDS/DQ3
IOT138B/LVDS/DQ3
IOT140A/LVDS/DQ3
IOT140B/LVDS/DQ3
IOT142A/LVDS/DQ3
IOT144A/LVDS/DQ3
IOT144B/LVDS/DQ3
IOT146A/TPLL1_T_IN1/TPLL1_T_FB0/LVDS/DQ3
U10B
GW5AT-LV60PG484A
Bank3
IOR5A/CCLK/LVDS
IOR5B/READY/LVDS
IOR7A/MODE1/LVDS
IOR7B/MODE0/LVDS
IOR9A/DONE/LVDS
IOR9B/RECONFIG_N/LVDS
IOR11A/TPLL1_T_IN0/TPLL1_T_FB1/LVDS
IOR11B/MODE2/TPLL1_C_IN0/TPLL1_C_FB1/LVDS
Bank4
IOR20A/LVDS/DQ4
IOR20B/LVDS/DQ4
IOR22A/LVDS/DQ4
IOR22B/LVDS/DQ4
IOR24A/LVDS/DQ4
IOR24B/LVDS/DQ4
IOR26A/LVDS/DQS4
IOR26B/LVDS/DQS4
IOR30A/LVDS/DQ4
IOR30B/LVDS/DQ4
IOR32A/LVDS/DQ4
IOR32B/LVDS/DQ4
IOR34A/LVDS/DQ4
IOR34B/LVDS/DQ4
IOR36A/RPLL0_T_IN1/RPLL0_C_FB0/LVDS/DQ4
IOR36B/RPLL0_C_IN1/RPLL0_C_FB0/LVDS/DQ4
U10C
FB10
BLM18KG121TN1D
FB11
BLM18KG121TN1D
C415
0.1uF
C416
0.1uF
C417
4.7uF
R112
10K
U12
NC
VCC
GND
OUT
50M_Active
R119
22R
CLK_50MHz
U11
OE
VDD
NC
CLKn
GND
CLKp
SiT9121-200M
C418
0.1uF
C419
0.1uF
PL_CLK_N
PL_CLK_P
R106
1K
MODE0
R107
DNI/1K
R108
DNI/1K
MODE1
R109
1K
R110
DNI/1K
MODE2
R111
1K
R113
4.7K
READY
R114
DNI/4.7K
R115
330R
DONE
R116
4.7K
D1
BULE
R117
4.7K
RECONFIG_N
R118
DNI/4.7K
GW5AT-LV60PG484A
Bank 6
IOB128A/D00/MOSI/MI0/LVDS/DQ6
IOB128B/D01/DIN/MISO/MI1/LVDS/DQ6
IOB130A/LVDS
IOB130B/LVDS
IOB132A/LVDS
IOB132B/LVDS
IOB134A/LVDS
IOB134B/LVDS
IOB136A/LVDS
IOB136B/LVDS
IOB138A/LVDS
IOB138B/LVDS
IOB140A/LVDS
IOB140B/LVDS
IOB142A/LVDS
IOB142B/LVDS
IOB144A/LVDS
IOB144B/LVDS
IOB146A/LVDS
IOB146B/LVDS
ADC
ADCTN
ADCTP
ADCVN
ADCVP
U10E
GW5AT-LV60PG484A
Bank 7
IOB102A/D11/LVDS/DQ7
IOB102B/D12/LVDS/DQ7
IOB104A/D09/LVDS/DQ7
IOB104B/D10/LVDS/DQ7
IOB106A/LVDS/DQ7
IOB106B/LVDS/DQ7
IOB108A/LVDS/DQ7
IOB108B/LVDS/DQ7
IOB111A/MCS_N/LVDS/DQ6
IOB111B/D08/LVDS/DQ6
IOB113A/D06/LVDS/DQ6
IOB113B/D07/LVDS/DQ6
IOB115A/D04/LVDS/DQ6
IOB115B/D05/LVDS/DQ6
IOB117A/PUDC_B/LVDS/DQS6
IOB117B/EMCCLK/LVDS/DQS6
IOB120A/D02/MI2/LVDS/DQ6
IOB120B/D03/MI3/LVDS/DQ6
IOB122A/LVDS/DQ6
IOB122B/LVDS/DQ6
IOB124A/LVDS/DQ6
IOB124B/LVDS/DQ6
IOB126A/LVDS/DQ6
IOB126B/LVDS/DQ6
U10F
R120
DNI/1K
R121
1K
GW5AT-LV60PG484A
Bank 5
IOR39A/RPLL0_T_IN0/RPLL0_T_FB1/LVDS
IOR39B/RPLL0_C_IN0/RPLL0_C_FB1/LVDS
IOR41A/GCLKT_7/LVDS
IOR41B/GCLKC_7/LVDS
IOR43A/LVDS
IOR43B/LVDS
IOR45A/LVDS
IOR45B/LVDS
IOR47A
IOR48A/LVDS
IOR48B/LVDS
IOR50A/LVDS
IOR50B/LVDS
IOR52A/LVDS
IOR52B/LVDS
IOR54A/GCLKT_5/LVDS
IOR54B/GCLKC_5/LVDS
IOR56A
IOR57A/GCLKT_6/LVDS/DQ5
IOR57B/GCLKC_6/LVDS/DQ5
IOR59A/LVDS/DQ5
IOR59B/LVDS/DQ5
IOR61A/SDA/GCLKT_4/RPLL1_T_IN2/RPLL1_T_FB0/LVDS/DQ5
IOR61B/SCL/GCLKC_4/RPLL1_C_IN2/RPLL1_C_FB0/LVDS/DQ5
IOR63A/SI/SSI0/LVDS/DQS5
IOR63B/SSPI_CS_N/LVDS/DQS5
IOR66A/SO/SSI1/RPLL1_T_IN1/RPLL1_T_FB1/LVDS/DQ5
IOR66B/SSPI_CLK/RPLL1_C_IN1/RPLL1_C_FB1/LVDS/DQ5
IOR68A/LVDS/DQ5
IOR68B/LVDS/DQ5
IOR70A/SSPI_WPN/SSI2/LVDS/DQ5
IOR70B/CLKHOLD_N/SSI3/LVDS/DQ5
IOR72A/RPLL1_T_IN0/LVDS/DQ5
IOR72B/RPLL1_C_IN0/LVDS/DQ5
U10D
R122
4.7K
R123
4.7K
C420
4.7uF
R124
4.7K
U13
nCS
VCC
DATA
nHOLD
nWP
DCLK
GND
ASDI
W25Q128JVSIQ
R125
4.7K
MSPI_CS_N
MSPI_MISO
MSPI_WP
MSPI_HOLD
MSPI_CLK
MSPI_MOSI
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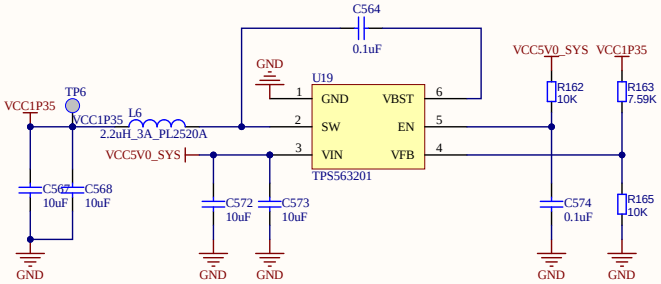

GW5AT-LV60PG484A
Bank 8
B8_B75_P U17 IOB75A/LVDS/DQ8
B8_B75_N U18 IOB75B/LVDS/DQ8
B8_B77_P AA18 IOB77A/BPLL1_T_IN2/BPLL1_C_FB0/LVDS/DQ8
B8_B77_N AB18 IOB77B/BPLL1_C_IN2/BPLL1_C_FB0/LVDS/DQ8
B8_B79_P V17 IOB79A/CSI_B/LVDS/DQ8
B8_B79_N W17 IOB79B/LVDS/DQ8
B8_B81_P V18 IOB81A/GCLKT_11/LVDS/DQ8
B8_B81_N V19 IOB81B/GCLKC_11/LVDS/DQ8
B8_B85_P AA19 IOB85A/RDWR_B/LVDS/DQS8
B8_B85_N AB20 IOB85B/CSO_B/DOUT/LVDS/DQS8
B8_B87_P Y18 IOB87A/GCLKT_10/LVDS/DQ8
B8_B87_N Y19 IOB87B/GCLKC_10/LVDS/DQ8
B8_B89_P AB15 IOB89A/LVDS/DQ8
B8_B89_N AA14 IOB89B/LVDS/DQ8
B8_B91_P W14 IOB91A/LVDS/DQ8
B8_B91_N V14 IOB91B/LVDS/DQ8
B8_B93_P W19 IOB93A/GCLKT_9/LVDS/DQ7
B8_B93_N W20 IOB93B/GCLKC_9/LVDS/DQ7
B8_B95_P U20 IOB95A/GCLKT_8/LVDS/DQ7
B8_B95_N V20 IOB95B/GCLKC_8/LVDS/DQ7
B8_B97_P AB21 IOB97A/D14/LVDS/DQ7
B8_B97_N AB22 IOB97B/D15/LVDS/DQ7
B8_B99_P Y21 IOB99A/LVDS/DQS7
B8_B99_N Y22 IOB99B/D13/LVDS/DQS7
U10G
GW5AT-LV60PG484A
Bank 9
AA9 IOB1A/LVDS/DQ12 IOB71B/BPLL1_C_IN1/BPLL1_C_FB1/LVDS/DQ9 V10 DDR_DQ15
AA10 IOB1B/LVDS/DQ12 IOB71A/BPLL1_T_IN1/BPLL1_C_FB1/LVDS/DQ9 U7
T3 IOB3A/BPLL0_T_IN1/BPLL0_T_FB0/LVDS/DQ12 IOB69B/LVDS/DQ9 Y9 DDR_DQ12
AB10 IOB3B/BPLL0_C_IN1/BPLL0_C_FB0/LVDS/DQ12 IOB69A/LVDS/DQ9 W9 DDR_DQ14
DDR_DQ27 T1 IOB5A/BPLL0_T_IN2/BPLL0_T_FB1/LVDS/DQ12 IOB67B/BPLL1_C_IN0/LVDS/DQ9 Y7 DDR_DQ8
DDR_DQ29 U1 IOB5B/BPLL0_C_IN2/BPLL0_C_FB1/LVDS/DQ12 IOB67A/BPLL1_T_IN0/LVDS/DQ9 Y8 DDR_DQ10
DDR_DQ30 U2 IOB7A/LVDS/DQ12 IOB65B/LVDS/DQ9 AB8 DDR_DQ9
DDR_DQ26 V2 IOB7B/LVDS/DQ12 IOB65A/LVDS/DQ9 AA8 DDR_DQ13
DDR_DQ25 W2 IOB9A/LVDS/DQ12 IOB62B/LVDS/DQS9 V8 DDR_DQS1_N
DDR_DQ24 Y2 IOB9B/LVDS/DQ12 IOB62A/LVDS/DQS9 V9 DDR_DQS1_P
DDR_DQS3_P R3 IOB12A/LVDS/DQS12 IOB60B/LVDS/DQ9 AB6 DDR_DM1
DDR_DQS3_N R2 IOB12B/LVDS/DQS12 IOB60A/LVDS/DQ9 AB7 DDR_DQ11
DDR_DQ31 W1 IOB14A/LVDS/DQ12 IOB58B/LVDS/DQ9 W7 DDR_CK_N
DDR_DM3 Y1 IOB14B/LVDS/DQ12 IOB58A/LVDS/DQ9 V7 DDR_CK_P
U3 IOB16A/LVDS/DQ12 IOB56B/LVDS/DQ10 Y14
DDR_DQ28 V3 IOB16B/LVDS/DQ12 IOB56A/LVDS/DQ9 Y13
AB12 IOB18A/LVDS/DQ11 IOB54B/LVDS/DQ10 AA13
AB11 IOB18B/LVDS/DQ11 IOB54A/LVDS/DQ10 AB13
AA11 IOB20A/LVDS/DQ11 IOB52B/LVDS/DQ10 W12
Y11 IOB20B/LVDS/DQ11 IOB52A/LVDS/DQ10 Y12
DDR_DQ16 AA1 IOB22A/LVDS/DQ11 IOB50B/LVDS/DQ10/DQS_910 AA6 DDR_DQ1
DDR_DQ20 AB1 IOB22B/LVDS/DQ11 IOB50A/LVDS/DQ10/DQS_910 Y6 DDR_DQ3
DDR_DQ18 AB3 IOB24A/LVDS/DQ11 IOB48B/LVDS/DQ10 T6 DDR_DQ5
DDR_DQ22 AB2 IOB24B/LVDS/DQ11 IOB48A/LVDS/DQ10 R6 DDR_DM0
DDR_DQS2_P Y3 IOB26A/LVDS/DQS11 IOB44B/LVDS/DQS10 W5 DDR_DQS0_N
DDR_DQS2_N AA3 IOB26B/LVDS/DQS11 IOB44A/LVDS/DQS10 W6 DDR_DQS0_P
DDR_DM2 AA5 IOB29A/LVDS/DQ11 IOB42B/LVDS/DQ10 V5 DDR_DQ7
DDR_DQ21 AB5 IOB29B/LVDS/DQ11 IOB42A/LVDS/DQ10 U6 DDR_DQ6
DDR_DQ17 Y4 IOB31A/GCLKT_14/LVDS/DQ11 IOB40B/GCLKC_12/LVDS/DQ10 U5 DDR_DQ4
DDR_DQ23 AA4 IOB31B/GCLKC_14/LVDS/DQ11 IOB40A/GCLKT_12/LVDS/DQ10 T5 DDR_DQ2
V4 IOB33A/GCLKT_15/LVDS/DQ11 IOB38B/GCLKC_13/LVDS/DQ10 T4 DDR_DQ0
DDR_DQ19 W4 IOB33B/GCLKC_15/LVDS/DQ11 IOB38A/GCLKT_13/LVDS/DQ10 R4
W11 IOB35A/LVDS/DQ10
W10 IOB35B/LVDS/DQ11
U10H
GW5AT-LV60PG484A
Bank 11
F4 IOL21B
B11_L23_P B1 IOL23A/LVDS
B11_L23_N A1 IOL23B/LVDS
B11_L25_P C2 IOL25A/LVDS
B11_L25_N B2 IOL25B/LVDS
B11_L27_P E1 IOL27A/LVDS
B11_L27_N D1 IOL27B/LVDS
B11_L29_P E2 IOL29A/LVDS
B11_L29_N D2 IOL29B/LVDS
B11_L31_P G1 IOL31A/LVDS
B11_L31_N F1 IOL31B/LVDS
B11_L33_P F3 IOL33A/LVDS
B11_L33_N E3 IOL33B/LVDS
B11_L35_P K1 IOL35A/LPLL0_T_IN0/LPLL0_T_FB0/LVDS
B11_L35_N J1 IOL35B/LPLL0_C_IN0/LPLL0_C_FB0/LVDS
B11_L37_P H2 IOL37A/LVDS
B11_L37_N G2 IOL37B/LVDS
B11_L39_P K2 IOL39A/LVDS
B11_L39_N J2 IOL39B/LVDS
B11_L41_P J5 IOL41A/LVDS
B11_L41_N H5 IOL41B/LVDS
B11_L43_P H3 IOL43A/GCLKT_19/LPLL0_T_IN1/LPLL0_T_FB1/LVDS
B11_L43_N G3 IOL43B/GCLKC_19/LPLL0_C_IN1/LPLL0_C_FB1/LVDS
B11_L45_P H4 IOL45A/GCLKT_18/LVDS
B11_L45_N G4 IOL45B/GCLKC_18/LVDS
Bank 12
FPGA_TCK V12 IOR1A/TCK
FPGA_TDI R13 IOR1B/TDI
FPGA_TMS T13 IOR3A/TMS
FPGA_TDO U13 IOR3B/TDO
U10J
GW5AT-LV60PG484A
Bank 10
DDR_A0 K4 IOL48A/GCLKT_17/LVDS
DDR_A7 J4 IOL48B/GCLKC_17/LVDS
DDR_A14 L3 IOL50A/GCLKT_16/LVDS
DDR_A3 K3 IOL50B/GCLKC_16/LVDS
DDR_A6 M1 IOL52A/LVDS
DDR_A8 L1 IOL52B/LVDS
DDR_A4 M3 IOL54A/LVDS
DDR_A11 M2 IOL54B/LVDS
DDR_A5 K6 IOL57A/LVDS
DDR_RESET J6 IOL57B/LVDS
DDR_ODT L5 IOL59A/LVDS
DDR_CS L4 IOL59B/LVDS
DDR_A13 N4 IOL61A/LVDS
DDR_BA1 N3 IOL61B/LVDS
DDR_CKE0 R1 IOL63A/LPLL1_T_IN0/LVDS
DDR_A12 P1 IOL63B/LPLL1_C_IN0/LVDS
DDR_BA0 L6 IOL65A
DDR_A2 P5 IOL66A/LPLL1_T_IN1/LPLL1_T_FB0/LVDS
DDR_A9 P4 IOL66B/LPLL1_C_IN1/LPLL1_C_FB0/LVDS
DDR_A10 P2 IOL68A/LPLL1_T_IN2/LPLL1_T_FB0/LVDS
DDR_A1 N2 IOL68B/LPLL1_C_IN2/LPLL1_C_FB0/LVDS
DDR_RAS M6 IOL70A/BPLL0_T_IN0/LVDS
DDR_CAS M5 IOL70B/BPLL0_C_IN0/LVDS
DDR_WE P6 IOL72A/LVDS
DDR_BA2 N5 IOL72B/LVDS
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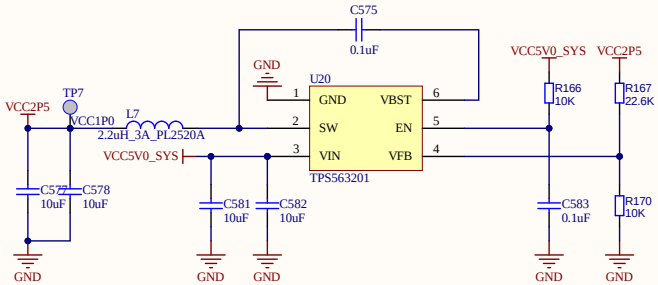


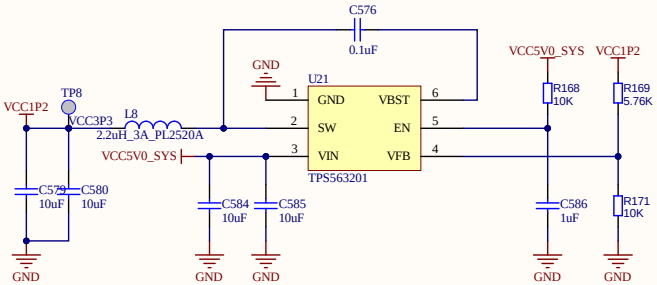
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MT41K128M16T-125
ADDR3
U16
MT41K128M16T-125
ADDR3
U17
TPS51200DRCR
VCC1P35
DDR_VREF
VCC3P3
DDR_VTT
GND
DDR_CK_P
DDR_CK_N
R131 100R
R132 DNI/2K
DDR_RESET
R128 240R/1%
R129 240R/1%
R130 4.7K
R133 10K
R134 100K
R138 10K
C517 0.01uF
C518 0.01uF
C519 0.01uF
C520 0.01uF
C521 4.7uF
C522 10uF
C523 10uF
C524 10uF
C525 10uF
C526 0.01uF
C527 4.7uF
C528 0.1uF
C529 4.7uF
C530 0.1uF
C531 0.1uF
C532 0.1uF
C533 0.1uF
C534 2.2nF
C535 2.2nF
C536 2.2nF
C537 2.2nF
C538 2.2nF
C539 10uF
C540 0.1uF
C541 0.1uF
C542 0.1uF
C543 0.1uF
C544 0.1uF
C545 2.2nF
C546 3.3nF
C547 4.7nF
C548 0.01uF
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C558 0.1uF
C559 0.1uF
C560 470nF
C561 470nF
C562 470nF
R135 51R DDR_CS
R136 51R DDR_CAS
R137 51R DDR_RAS
R139 51R DDR_ODT
R140 51R DDR_A12
R141 51R DDR_BA2
R142 51R DDR_A4
R143 51R DDR_A3
R144 51R DDR_BA0
R145 51R DDR_BA1
R146 51R DDR_WE
R147 51R DDR_A10
R148 51R DDR_A2
R149 51R DDR_A5
R150 51R DDR_A0
R151 51R DDR_A1
R152 51R DDR_A13
R153 51R DDR_A7
R154 51R DDR_A8
R155 51R DDR_A9
R156 51R DDR_A6
R157 51R DDR_A11
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R159 51R DDR_CKE0
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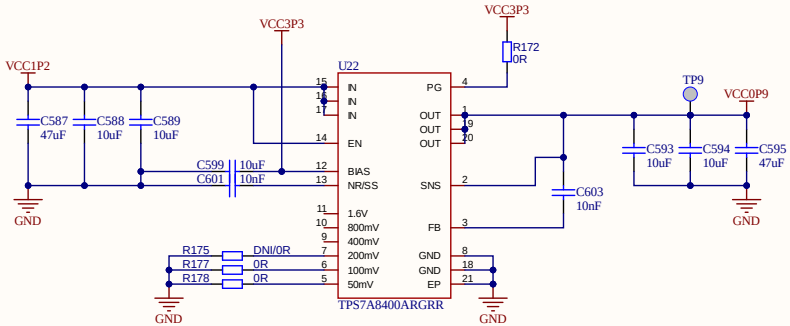


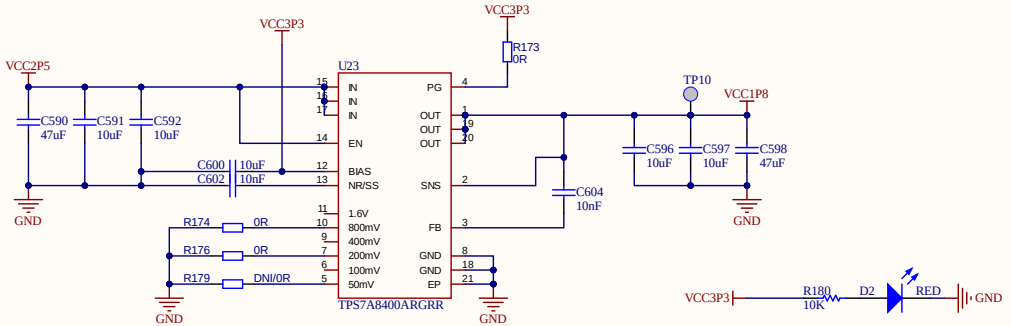


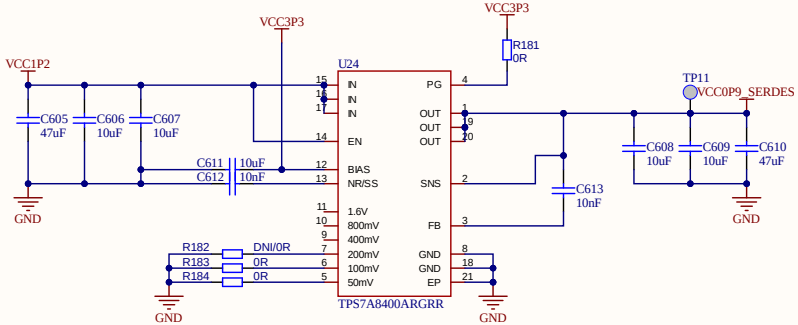




$$Vout = 0.768 * (1+ R1/R2)$$

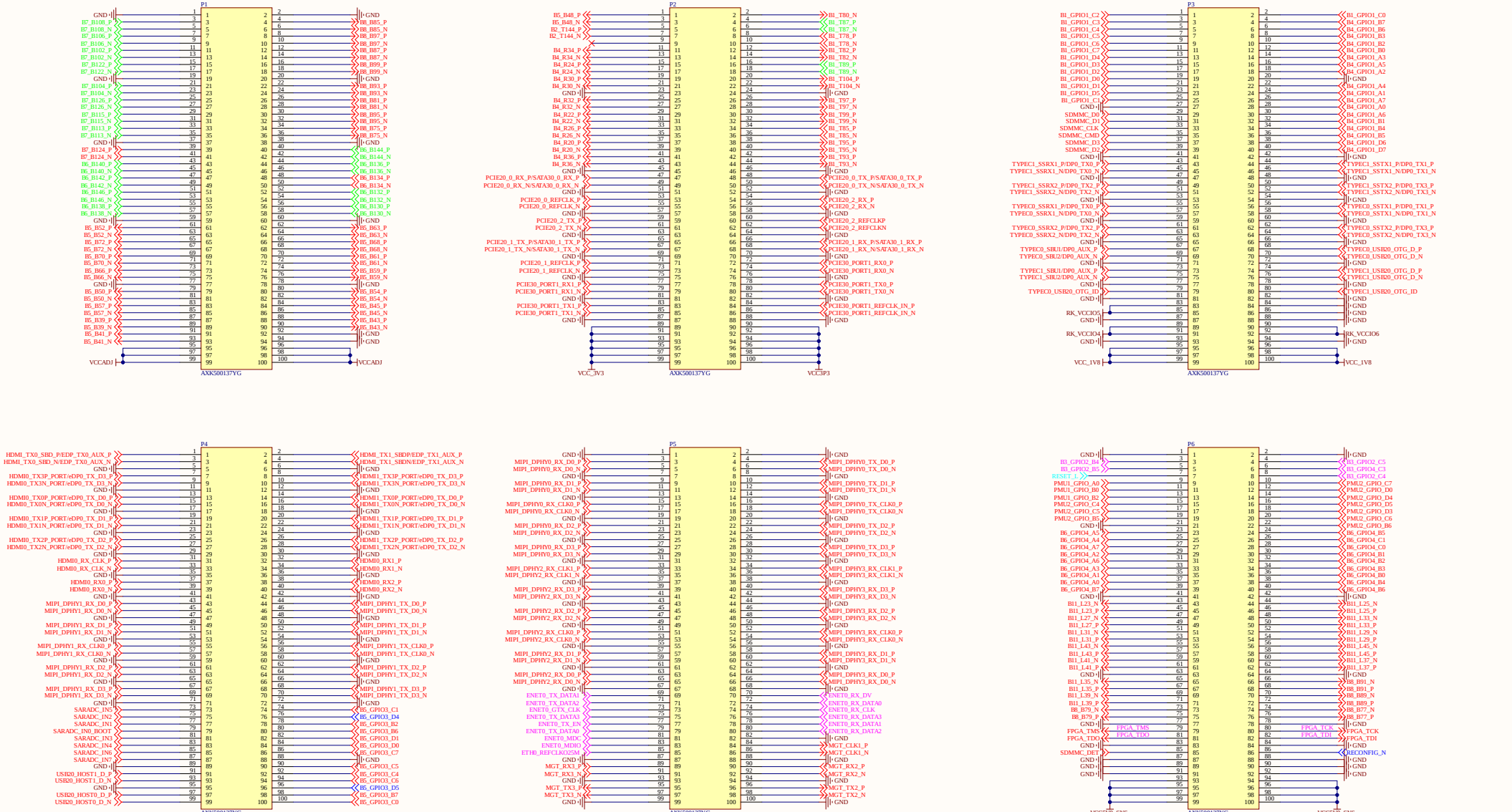






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FMC-LPC



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